

Price Volatility and Investor Behavior in an Overlapping Generations Model with Information Asymmetry

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ABSTRACT

This paper studies an overlapping generations model with multiple securities and heterogeneously informed agents. The model produces multiple equilibria, including highly volatile equilibria that can exhibit strong or weak correlations between asset returns—even when asset supplies and future dividends are uncorrelated across assets. Less informed agents rationally behave like trend-followers, while better informed agents follow contrarian strategies. Trading volume has a hump-shaped relation with information precision and is positively correlated with absolute price changes. Finally, accurate information increases the volatility and correlation of stock returns in the highly volatile, strongly correlated equilibrium.

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There is mounting evidence of both trend-following and contrarian behavior among various investor groups in recent empirical studies. Trend-followers buy assets upon price appreciation and sell them upon depreciation, while contrarians trade in the opposite way. Such trading behavior is found in both domestic and international markets. Moreover, prices in these markets are found to vary much more than the stocks' fundamental values. Indeed in some markets, prices exhibit common movements that are hard to explain by movements in the fundamentals.¹

Several strands of the literature have tried to reconcile these empirical findings with the theory. Prominent among them are overlapping generations models. Using a multiple-security model, Spiegel (1998) demonstrates the existence of a highly volatile equilibrium in which small supply shocks produce disproportionately large price variances. Spiegel shows that this can occur in an economy populated by overlapping generations of rational competitive agents. However, his agents are homogeneously informed and hence there is no heterogeneity in their trading patterns.

This paper builds on Spiegel's (1998) work by incorporating heterogeneous information with possibly asymmetric information precision among agents. This allows us to analyze differential trading behavior across various investor classes documented in the empirical literature while maintaining the qualitative nature of Spiegel's (1998) primary conclusions on excess volatility. The model is characterized by a multiple-security economy with overlapping generations of heterogeneously informed agents. Risky claims (stocks) on a single consumption good are traded in financial markets. A continuum of rational risk-averse agents lives for two periods. Upon birth, the agents receive noisy private signals about one-period-ahead dividends. Based on their private signals, their random endowments of the risky assets, and market prices, the agents make their investment decisions. When old, they unwind their security positions, consume, and die. The economy is then run by the next generation. Because stocks are in random supply, their prices reveal future dividends only partially and therefore serve as noisy public signals about the stocks' fundamentals. Thus, the model also belongs to the noisy rational expectations literature pioneered by Hellwig (1980) and Diamond and Verrecchia (1981), and later developed by Admati (1985) and others.² Our model can be considered an extension of Spiegel's (1998) model to a noisy rational expectations equilibrium framework, or of Admati's (1985) model to an overlapping generations economy.

¹For evidence of trend-following and contrarian behavior in domestic markets, see, for example, Bange (2000), Chordia, Roll, and Subrahmanyam (2002), and Goetzmann and Massa (2002, 2003). International evidence is documented in Bohn and Tesar (1996), Brennan and Cao (1997), Choe, Kho, and Stulz (1999), Froot, O'Connell, and Seasholes (2001), and Grinblatt and Keloharju (2000, 2001). Among the most cited evidence on excessive volatility are LeRoy and Porter (1981), Shiller (1981a, 1981b, 1989b), and West (1988). Excess comovement is the primary subject of Barberis, Shleifer, and Wurgler (2005), Morck, Yeung, and Yu (2000), Pindyck and Rotemberg (1990), and Shiller (1989a), and is also found in the prices of closed-end funds (Lee, Shleifer, and Thaler (1991)), closed-end country funds (Bodurtha, Kim, and Lee (1995)), and the S&P 500 component stocks (Vijh (1994)).

²For subsequent work, see, for example, Brown and Jennings (1989), Grundy and McNichols (1989), Kim and Verrecchia (1991a, 1991b), Brennan and Cao (1996, 1997), Cao (1999), Grundy and Kim (2002), and Kodres and Pritsker (2002).

As is often the case with an overlapping generations model, the model produces multiple equilibria. Specifically, as in Spiegel (1998) there potentially exist 2^K equilibria when K securities trade. These equilibria include highly volatile equilibria that can exhibit strong or weak cross-sectional correlations between changes in individual stock prices. Strikingly, this is true even when asset supplies and future dividends are uncorrelated across assets. Other equilibria include a low volatility equilibrium in which the volatility and correlation of price changes are of comparable magnitude to those of dividends. While multiplicity of partially revealing equilibria is not uncommon in noisy rational expectations equilibrium models, it is primarily due to self-fulfilling prophecies of overlapping generations in our model, as we demonstrate the existence of multiple equilibria even when agents have full or no information.

A partially revealing equilibrium allows us to analyze the effects that heterogeneous information has on prices and trades. We find that the volatility of changes in individual stock prices increases with information quality in a high volatility equilibrium, while it falls in a low volatility equilibrium. Similarly, the cross-sectional correlation between price changes becomes stronger with information quality in a high correlation equilibrium. This is true even when all agents have the same degree of information precision and hence there is no adverse selection problem. When there is information asymmetry among agents, less informed agents tend to purchase securities upon price appreciation, while better informed agents sell them. That is, less informed investors behave like trend-followers, while better informed investors follow profitable contrarian strategies. The intuition here is similar to Brennan and Cao (1996, 1997) and Kim and Verrecchia (1991b). With poorer private information, less informed agents rely more heavily on public price signals and therefore trade in the same direction as price changes. Under the setting considered in this paper, accurate information weakens agents' trend-following and contrarian behavior since it alleviates information asymmetry.³

Under partial revelation, trading volume is strictly positive and has a hump-shaped relation with average information accuracy. This arises because agents are effectively homogeneously informed or uninformed in the two extreme cases of full and no information; in these cases there is no informational motive to trade, and the volume is lower than it is with partial information. In addition, absolute trade flows are positively correlated with absolute price changes, consistent with the empirical evidence in Karpoff (1987) and Gallant, Rossi, and Tauchen (1992). The positive correlation weakens as private information becomes more precise on average. Of course, we are not the first to show these results; for example, results similar to the hump-shaped relation between volume and information precision also hold in Blume, Easley, and O'Hara (1994, Figure 1) and Holden and Subrahmanyam (2002, Proposition 1), and Wang (1994, Section V) finds a positive relationship between volume and absolute price changes. The current article complements these works by showing that the results above can also occur in highly

³The trend-following and contrarian behavior in this paper results from purely informational motives and should be distinguished from such behavior due to behavioral motives discussed in Barberis, Shleifer, and Vishny (1998), Daniel, Hirshleifer, and Subrahmanyam (1998), De Long et al. (1990b), and Hong and Stein (1999).

volatile, strongly correlated markets. We demonstrate these points by calibrating the model with parameter values estimated in the empirical literature. It is shown that for any level of information accuracy, only very small supply shocks are necessary to produce the observed levels of stock price volatility and correlation.

The key ingredients of the current model, namely, overlapping generations and heterogeneous information, are two major workhorses in addressing excess volatility and investor behavior. Using an overlapping generations model, De Long et al. (1990a) show that unpredictability of noise traders' erroneous beliefs prevents rational arbitrageurs from stabilizing price variability.⁴ Incorporating costly stock market participation in an overlapping generations model, Orosel (1998) demonstrates the occurrence of "rational trend chasing" by way of increased participation and return volatility. Neither of these papers, however, examines the effect of information on trades. In the information literature, Campbell and Kyle (1993) show that the interaction between rational "smart money" investors and exogenous noise traders can produce volatility levels that are consistent with the data. Wang (1993) demonstrates that asymmetric information, along with supply shocks, can increase price variability and that less informed agents may rationally behave like price chasers. Such trend-following behavior also occurs in the noisy rational expectations equilibrium models of Brennan and Cao (1996, 1997) and Kim and Verrecchia (1991b). In a model with multiple classes of investors who observe signals about either the payoff or supply of an asset, Gennotte and Leland (1990) show how changes in supply, caused by random liquidity trading and deterministic hedging plans, can dramatically affect market liquidity and price volatility. A distinguishing feature of our model from those of these authors is that very small supply shocks can dominate dividend shocks in equilibrium prices and become a major component of the second moments of returns.⁵ In addition, none of the studies cited here investigates comovement of asset prices.⁶ Finally, to the best of my knowledge, at least one paper incorporates both of the two key ingredients discussed in this paragraph. Biais, Bossaerts, and Spatt (2006) analyze the properties of a noisy rational expectations equilibrium with overlapping generations of informed and uninformed investors. Their main objective is to examine the implications of information asymmetry on asset pricing and investors' portfolio decisions. In contrast, our focus is on the analysis of the second moments of asset returns and the trading behavior of heterogeneously informed agents. In this sense, the current paper is complementary to theirs.

⁴Bhushan, Brown, and Mello (1997) demonstrate that myopia of traders is neither a necessary nor a sufficient condition for prices to be noisy in a setting such as De Long et al. (1990a).

⁵Coval (2000) also develops an asymmetric information model with random supply. Like our partially revealing equilibrium, the model is numerically solved to produce multiple equilibria with differential volatility levels. As he notes, however, "as many of the results are qualitatively similar for the two equilibria, [he] focus[es] on the low volatility equilibrium." (Section 4.1) In contrast, the high and low volatility equilibria in the current model have opposing return characteristics, and it is in the former that we are primarily interested.

⁶Outside the two categories discussed here, Barberis and Shleifer (2003) and Barberis, Shleifer, and Wurgler (2005) develop models of comovement. However, their focus is on the category and habitat views of comovement, as opposed to the rational one considered in the current paper.

The paper is organized as follows. The next section develops the model, solves for an equilibrium, and presents analytic results under full and no information. Section II examines the properties of partial-information equilibria and investigates trading behavior of asymmetrically informed investors. The final section, Section III, concludes and explores future agenda. The Appendix contains all proofs.

I. Overlapping Generations Model with Heterogeneous Information

A. Setup

The model extends Spiegel (1998) to a setting with heterogeneous information. The economy is populated by a continuum of rational risk-averse agents who consume a single good. There are K risky assets, called stocks, and a riskless bond available for trading in financial markets. Both types of securities pay in units of the consumption good. The dividend and supply processes of stocks follow random walks. At the beginning of period t , the stocks pay a vector of stochastic dividends $\tilde{D}_t$ per share, where

$$\tilde{D}_t = \tilde{D}_{t-1} + \tilde{\delta}_t. \quad (1)$$

The vector of dividend shocks, $\tilde{\delta}_t$, is distributed multivariate normal with zero mean and covariance matrix Σ_δ . Assuming zero mean is innocuous since we are primarily interested in the second moments of observable quantities. The Σ_δ matrix and all other variance-covariance matrices to be introduced are assumed positive definite unless otherwise noted.⁷

Per capita supply of stocks, $\tilde{N}_t$, is stochastic and also follows a vector random walk process⁸

$$\tilde{N}_t = \tilde{N}_{t-1} + \tilde{\eta}_t. \quad (2)$$

Again, the vector of unobservable supply shocks, $\tilde{\eta}_t$, is distributed multivariate normal with zero mean and covariance matrix Σ_η .⁹ The riskless bond pays r units of the consumption good as interest at the beginning of each period. It serves as numeraire for the economy and thus always sells for a price of unity. The gross interest rate is denoted by $R = 1 + r$. For stock prices to be finite, we require that $r > 0$.

⁷We require positive definiteness for a partial-information equilibrium to be well defined. An equilibrium can still exist when some covariance matrices are positive semidefinite, for example, $\Sigma_\delta = 0$. We use such a sure-dividend example below to derive intuition about full-information equilibria.

⁸The model can be extended in a straightforward manner to accommodate mean-reverting dividend and supply processes, $\tilde{D}_t = \alpha_D \tilde{D}_{t-1} + \tilde{\delta}_t$ and $\tilde{N}_t = \alpha_N \tilde{N}_{t-1} + \tilde{\eta}_t$, $-1 < \alpha_D, \alpha_N \leq 1$. For ease of exposition, we focus on the random walk specification.

⁹The supply of risky assets can become random through a variety of mechanisms, such as creation or destruction of the capital base in the economy and liquidity trading. For a discussion of possibly different empirical implications among these mechanisms, see Spiegel (1999).

Agents live for two periods while the economy goes on forever. In each period, a new generation of agents is born. There is a continuum of agents with unit mass, each of whom acts competitively taking prices as given. An agent, indexed by $i \in [0, 1]$, possesses negative exponential utility with constant absolute risk aversion (CARA) θ_i . The agent comes endowed with units of the bond and a personal share of supply shocks. The stock endowment is given by¹⁰

$$\tilde{\eta}_{t,i} = \tilde{\eta}_t + \tilde{\zeta}_{t,i}. \quad (3)$$

The noise component, $\tilde{\zeta}_{t,i}$, is unobservable, independent across agents, and distributed multivariate normal with zero mean and covariance matrix Σ_{ζ} .¹¹ This ensures that per capita endowment equals per capita supply shocks, or that $\int_i \tilde{\eta}_{t,i} di = \tilde{\eta}_t$ almost surely by the law of large numbers. The above formulation implies that knowing his own endowment provides an agent with some information about the aggregate supply shocks, which he takes into account in making portfolio decisions.¹²

The information structure is similar to that in Admati (1985). Upon birth, agent i receives a vector of noisy private signals about the one-period-ahead dividends,

$$\tilde{z}_{t,i} = \tilde{\delta}_{t+1} + \tilde{\varepsilon}_{t,i}.$$

The vector of unobservable noises, $\tilde{\varepsilon}_{t,i}$, is distributed multivariate normal with zero mean and covariance matrix $\Sigma_{\varepsilon,i}$. The $\tilde{\varepsilon}_{t,i}$'s are independent across agents, implying that the agents are heterogeneously informed. Information *accuracy*, however, can be either heterogeneous ($\Sigma_{\varepsilon,i} \neq \Sigma_{\varepsilon,j}, \exists i, j$) or homogeneous ($\Sigma_{\varepsilon,i} \equiv \Sigma_{\varepsilon}, \forall i$). When private signals are infinitely noisy, they reveal no information about the future dividends. This case corresponds to Spiegel's (1998) model. At the other extreme, when the $\tilde{\varepsilon}_{t,i}$'s have zero variance, private signals perfectly reveal $\tilde{\delta}_{t+1}$. In intermediate cases, the signals reveal only partial information about future dividends. For convenience, we refer to these three cases as the no-, full-, and partial-information models, respectively.¹³ We also refer to both of the first two cases as homogeneous-information models, since in these cases agents are homogeneously uninformed or informed. It is assumed that $\tilde{\delta}_t$, $\tilde{\eta}_t$, $\tilde{\zeta}_{t,i}$, and $\tilde{\varepsilon}_{t,i} \forall i$ are mutually and serially independent.

After the stocks and the bond pay their owners at the beginning of period t , trading takes place. As in Spiegel (1998), agents observe current prices ($\tilde{P}_t$) and dividends ($\tilde{D}_t$), and the whole history of past

¹⁰We do not specify the bond endowment since it does not affect the equilibrium in any way. See footnote 15.

¹¹It is straightforward to extend the model to a setting with non-identically distributed endowment noises, that is, $\tilde{\zeta}_{t,i} \sim N(0, \Sigma_{\zeta,i})$. For brevity, we make the *i.i.d.* assumption.

¹²The information content of random endowments often is made null (Grundy and McNichols (1989, p.498)) or is ignored (Brown and Jennings (1989, footnote 3)) in a large or continuum-of-agents economy. As Blume, Easley, and O'Hara (1994) discuss, however, the former approach produces infinite trading volume in the period when the random endowments are introduced. Our setting avoids this issue without ignoring the information content of endowments. For a *finite*-economy model that explicitly considers this information, see Diamond and Verrecchia (1981). Gennotte and Leland (1990) also introduce a class of competitive investors who observe a common signal about supply shocks created by liquidity traders.

¹³The corresponding equilibria are referenced analogously. We may also call the full- and partial-information equilibria the fully and partially revealing equilibria, respectively, in accordance with the terminology in the noisy rational expectations equilibrium literature.

prices, realized dividends, and supply levels. In addition, they use private signals $(\tilde{z}_{t,i})$ and individual endowments $(\tilde{\eta}_{t,i})$ to make their portfolio decisions. As we show below, under homogeneous information this implies that while agents do not observe current supply $(\tilde{N}_t)$, they can deduce it from market prices even though it is a priori unknown. When old at the beginning of period $t + 1$, they receive dividends from their portfolios, unwind their security positions, consume, and die. The economy is then run by the generation $t + 1$ agents and the whole cycle repeats.¹⁴

The next subsection begins the analysis by solving for an equilibrium.

B. Equilibrium

We focus on a linear equilibrium in which the price function takes the general form

$$\tilde{P}_t = A_1 \tilde{N}_{t-1} + A_2 \tilde{\eta}_t + B_1 \tilde{D}_t + B_2 \tilde{\delta}_{t+1} + c, \quad (4)$$

where A_1 , A_2 , B_1 , and B_2 are K -dimensional square matrices and c is a K -dimensional vector to be determined. We only look for a stationary equilibrium in which these coefficients are time invariant.

Let $X_{t,i}$ be agent i 's stock holdings in period t . His future wealth, $\tilde{W}_{t+1,i}$, is then given by

$$\tilde{W}_{t+1,i} = X_{t,i}^T \tilde{Q}_{t+1} + R W_{t,i}, \quad (5)$$

$$\tilde{Q}_{t+1} \equiv \tilde{P}_{t+1} + \tilde{D}_{t+1} - R \tilde{P}_t, \quad (6)$$

where $\tilde{Q}_{t+1}$ is the vector of excess returns per share and $W_{t,i}$ is the agent's exogenously given endowment.¹⁵ Notice that even under full information with perfect knowledge about future dividends $\tilde{D}_{t+1}$, future wealth still remains uncertain because the one-period-ahead prices depend on yet unknown $\tilde{\delta}_{t+2}$ and $\tilde{\eta}_{t+1}$ given the price conjecture in (4). Thus, the utility maximization problem is always well defined. Since all stochastic variables are distributed multivariate normal, $\tilde{W}_{t+1,i}$ is (univariate) normally distributed. Let $\mathcal{F}_{t,i} = \{\tilde{z}_{t,i}, \tilde{\eta}_{t,i}, \tilde{P}_t, \tilde{D}_t, \tilde{N}_{t-1}\}$ denote agent i 's information set.¹⁶ By the property of negative exponential utility, agent i 's optimization problem, $\max_{X_{t,i}} \mathbf{E}[-\exp(-\theta_i \tilde{W}_{t+1,i}) | \mathcal{F}_{t,i}]$, amounts to maximizing the certainty equivalent of future wealth:

$$\max_{X_{t,i}} \mathbf{E}[\tilde{W}_{t+1,i} | \mathcal{F}_{t,i}] - \frac{\theta_i}{2} \text{Var}[\tilde{W}_{t+1,i} | \mathcal{F}_{t,i}]. \quad (7)$$

¹⁴The terms "generation" and "period" are used interchangeably hereafter.

¹⁵To derive these expressions, let $b_{t,i}$ denote agent i 's bond holdings. Then, $\tilde{W}_{t+1,i} = X_{t,i}^T (\tilde{P}_{t+1} + \tilde{D}_{t+1}) + b_{t,i} R$. His budget constraint is given by $X_{t,i}^T \tilde{P}_t + b_{t,i} = W_{t,i}$. Eliminating $b_{t,i}$ from these two equations gives the expressions in the text. The endowment $W_{t,i}$ equals the value of the stock endowment, $\tilde{\eta}_{t,i}^T \tilde{P}_t$, plus the number of endowed bonds. The bond endowment does not affect the equilibrium stock holdings because it drops out from the first-order condition due to the CARA utility assumption.

¹⁶This is the full information set under partial revelation. In a full- or no-information model, the information set is identical across agents, and some of its members shown here are redundant. We keep the i subscript for notational consistency with the partial-information model. We also keep the tilde above the variables for the same reason even when some quantities may be known.

The first-order condition is given by

$$X_{t,i} = \frac{1}{\theta_i} \text{Var}^{-1}(\tilde{Q}_{t+1}|\mathcal{F}_{t,i}) \mathbf{E}[\tilde{Q}_{t+1}|\mathcal{F}_{t,i}]. \quad (8)$$

The second-order condition for maximization is met if $\text{Var}(\tilde{Q}_{t+1}|\mathcal{F}_{t,i})$ is positive definite. As usual, the equilibrium condition is that per capita demand equals per capita supply,

$$\int_i X_{t,i} di = \tilde{N}_t. \quad (9)$$

Comparing both sides of this equation determines the price coefficients in (4). The following theorem summarizes the result.

THEOREM 1 (Equilibrium): *An equilibrium at the respective information level is characterized by the following price function, $\tilde{P}_t$, and the demand function $X_{t,i}$:*

(i) *Full information:* $A_1 = A_2 \equiv A$, $B_1 = B_2 = \frac{1}{r}I$, and $c = 0$ in equation (4). Specifically,

$$\tilde{P}_t = A\tilde{N}_t + \frac{1}{r}\tilde{D}_{t+1}, \quad (10)$$

where A is a symmetric negative-definite matrix that satisfies the quadratic matrix equation

$$A\Sigma_\eta A + \frac{r}{\theta}A + \frac{1}{r^2}\Sigma_\delta = 0. \quad (11)$$

(ii) *No information (Spiegel (1998)):* $A_1 = A_2 \equiv A$, $B_1 = \frac{1}{r}I$, $B_2 = 0$, and $c = 0$ in equation (4). That is,

$$\tilde{P}_t = A\tilde{N}_t + \frac{1}{r}\tilde{D}_t, \quad (12)$$

where A is a symmetric negative-definite matrix that satisfies the quadratic matrix equation

$$A\Sigma_\eta A + \frac{r}{\theta}A + \frac{R^2}{r^2}\Sigma_\delta = 0. \quad (13)$$

The demand function under full or no information is given by

$$X_{t,i} = \frac{\bar{\theta}}{\theta_i} \tilde{N}_t, \quad (14)$$

where $\bar{\theta}$ is the harmonic mean of individual risk-aversion parameters, $\bar{\theta} = (\int_i \theta_i^{-1} di)^{-1}$.

(iii) *Partial information:* Generally, $A_1 \neq A_2$, $B_1 = \frac{1}{r}I \neq B_2$, and $c = 0$ in equation (4). In particular,

$$\tilde{P}_t = A_1\tilde{N}_{t-1} + \frac{1}{r}\tilde{D}_t + B_2\tilde{\xi}_t, \quad (15)$$

where $\tilde{\xi}_t \equiv \tilde{\delta}_{t+1} + B_2^{-1}A_2\tilde{\eta}_t$ and A_1 , A_2 , and B_2 are nonsingular matrices that solve a system of nonlinear matrix equations given in the Appendix. In addition, A_1 is symmetric and negative definite. The demand function is linear in $\tilde{\xi}_t$, $\tilde{z}_{t,i}$, $\tilde{\eta}_{t,i}$, $\tilde{D}_t$, and $\tilde{N}_{t-1}$.

Proof. The Appendix contains all proofs. ■

We first analyze the homogeneous-information equilibria for which closed-form solutions are available. This provides useful insights into the analysis of partial-information equilibria in Section II. Equation (10) says that prices under full information are the present value of a perpetuity paying $\tilde{D}_{t+1}$ less a discount due to supply pressure, $A\tilde{N}_t$. Prices depend on one-period-ahead dividends since they are perfectly forecastable. The $A\tilde{N}_t$ term is a “discount” if stocks are in positive supply since A is negative definite. The price function (12) under no information takes a similar form, but the perpetuity consists only of current dividends $\tilde{D}_t$ because agents have no information about future dividends. Due to informational homogeneity, the demand function in (14) merely reflects the market-making activity of competitive agents who simply accommodate supply shocks inversely with their risk aversion. The demand function also implies that two-fund monetary separation holds under homogeneous information; each agent holds a combination of the market portfolio, $\tilde{N}_t$, and the bond. As one might anticipate from the normality assumption and homogeneous expectations, a version of the Capital Asset Pricing Model (CAPM) holds, with dividend shocks augmented by supply shocks.

The quadratic matrix equations (11) and (13) are easy to interpret: They are simply the market-clearing conditions. To see this, substitute the first-order condition (8) into equation (9) and rearrange to obtain

$$Var(\tilde{Q}_{t+1}|\mathcal{F}_{t,i})\tilde{N}_t = \mathbf{E}[\tilde{Q}_{t+1}|\mathcal{F}_{t,i}]/\bar{\theta}, \quad (16)$$

where we note that the conditional expectation and variance here are identical across agents. This expression says that the risk of holding stocks in the left-hand side must be compensated by expected returns per unit average risk aversion in the right-hand side. Given the price function (10), in a full-information equilibrium the variance on the left-hand side is $A\Sigma_\eta A + \Sigma_\delta/r^2$. In a no-information equilibrium this is $A\Sigma_\eta A + \Sigma_\delta R^2/r^2$, that is, the lack of knowledge about future dividends increases the dividend portion of the variance (the second term) by a factor of $R^2 = (1+r)^2$. The expected return on the right-hand side is the “net return” on the price discount, $-rA\tilde{N}_t$. Equating the coefficients on $\tilde{N}_t$ yields the respective matrix equations.

The A matrix, the sensitivity of prices to supply shocks, plays an important role in determining equilibrium characteristics. It prescribes how the supply shock of a stock affects its own price and, if nondiagonal, the prices of other stocks as well. Since the A matrix is determined by a quadratic equation, one may well expect the existence of multiple equilibria. The following corollary shows that this is indeed the case.

COROLLARY 1 (*Solutions under homogeneous information*): *The analytic solutions to the quadratic matrix equations (11) and (13) are both given in the form*

$$A = -\frac{r}{2\bar{\theta}}\Sigma_\eta^{-1} + \Sigma_\eta^{-\frac{1}{2}}C\Lambda_{\pm}^{\frac{1}{2}}C^T\Sigma_\eta^{-\frac{1}{2}}, \quad (17)$$

where $\Sigma_\eta^{\frac{1}{2}}$ is the unique symmetric positive-definite square root of Σ_η , C and Λ are the matrices of orthonormal eigenvectors and eigenvalues, respectively, of

$$\begin{aligned} M_{FI} &\equiv \frac{r^2}{4\bar{\theta}^2} I - \frac{1}{r^2} \Sigma_\eta^{\frac{1}{2}} \Sigma_\delta \Sigma_\eta^{\frac{1}{2}} \text{ for (11), or} \\ M_{NI} &\equiv \frac{r^2}{4\bar{\theta}^2} I - \frac{R^2}{r^2} \Sigma_\eta^{\frac{1}{2}} \Sigma_\delta \Sigma_\eta^{\frac{1}{2}} \text{ for (13),} \end{aligned} \quad (18)$$

I is the identity matrix, and $\Lambda_\pm^{\frac{1}{2}}$ denotes a matrix obtained by taking the square roots of the diagonal elements of Λ and changing their signs freely.

The A matrix is real-valued if and only if the corresponding M matrix above is positive semidefinite. To have strictly multiple equilibria, we assume that M_{NI} , and hence M_{FI} , are positive definite for the rest of the paper. This is likely the case when, ceteris paribus, future cash flows are discounted enough (r is high), agents are risk tolerant ($\bar{\theta}$ is small), and the dividend- and supply-shock variances are small (Σ_η and Σ_δ are “small” in some matrix norm). Each equilibrium corresponds to a different value of $\Lambda_\pm^{\frac{1}{2}}$.¹⁷ In total there are 2^K equilibria when K securities trade. Economically, the eigenvector matrix C controls the cross-sectional dependency of supply shocks, and the signed square roots of the eigenvalues in $\Lambda_\pm^{\frac{1}{2}}$ determine the price sensitivity to supply shocks.

C. Price Volatility and Correlation in Homogeneous-Information Equilibria

Using the equilibrium characterization obtained in the previous subsection, we now study the volatility and correlation of changes in asset prices under homogeneous information. From Theorem 1, the vector of price changes under full information is given by¹⁸

$$\Delta \tilde{P}_t \equiv \tilde{P}_t - \tilde{P}_{t-1} = A \tilde{\eta}_t + \frac{1}{r} \tilde{\delta}_{t+1}.$$

Thus, the variance of the price changes is

$$Var_{FI}(\Delta \tilde{P}_t) = A \Sigma_\eta A + \frac{1}{r^2} \Sigma_\delta = -\frac{r}{\bar{\theta}} A, \quad (19)$$

where we use the quadratic matrix equation (11) in the second equality. Similarly, we can show that the variance of price changes in a no-information equilibrium is given by¹⁹

$$Var_{NI}(\Delta \tilde{P}_t) = -\frac{r}{\bar{\theta}} A - \frac{R^2 - 1}{r^2} \Sigma_\delta. \quad (20)$$

That is, under homogeneous information the variance of price changes is linear in A , the price sensitivity to supply shocks. From equations (19) and (20), it is easy to see that the volatility and correlation of

¹⁷The sign $\pm$ is used to signify nonuniqueness.

¹⁸Since the vector of cum-dividend price changes, $\tilde{P}_t + \tilde{D}_t - \tilde{P}_{t-1}$, is nonstationary due to the random-walk assumption, we work with ex-dividend price changes, $\tilde{P}_t - \tilde{P}_{t-1}$.

¹⁹Note that the A matrix has different values between the two equilibria and therefore we cannot directly compare the two variance formulae. The variance levels between equilibria with differential information precisions will be discussed below.

price changes will differ across equilibria corresponding to different values of A . We first establish the following proposition about volatility, which is similar to Spiegel (1998, Proposition 3):

PROPOSITION 1 (*Variance of price changes under homogeneous information*): *Consider switching between two equilibria under full or no information by changing the sign of any diagonal element of $\Lambda_{\pm}^{\frac{1}{2}}$ in (17). Switching the sign from positive to negative increases the variance of the change in almost any portfolio's value.*

Unlike volatility, the cross-sectional correlation between price changes is difficult to analyze without specifying the form of the underlying shock-covariance matrices. If one has no prior knowledge about the securities in the economy, it seems natural to assume that stocks are cross-sectionally symmetric in their underlying shocks, as formalized below.

ASSUMPTION 1 (*Symmetric securities*): *There are $K \geq 2$ securities with cross-sectionally identical dividend- and supply-shock variances as well as correlations,*

$$\Sigma_{\delta} = \sigma_{\delta}^2 \begin{pmatrix} 1 & \rho_{\delta} & \cdots & \rho_{\delta} \\ \rho_{\delta} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho_{\delta} \\ \rho_{\delta} & \cdots & \rho_{\delta} & 1 \end{pmatrix}, \quad \Sigma_{\eta} = \sigma_{\eta}^2 \begin{pmatrix} 1 & \rho_{\eta} & \cdots & \rho_{\eta} \\ \rho_{\eta} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho_{\eta} \\ \rho_{\eta} & \cdots & \rho_{\eta} & 1 \end{pmatrix}.$$

Under this assumption, we can say much about the properties of the equilibria.²⁰ As stated before, there exist 2^K equilibria when K securities trade (if the M matrices in Corollary 1 are positive definite). In some of these equilibria, stocks have asymmetric price properties depending on the choice of $\Lambda_{\pm}^{\frac{1}{2}}$ in Corollary 1, even though the distributions of the underlying shocks are symmetric. Throughout the paper, however, we focus on the following four symmetric equilibria:

PROPOSITION 2 (*Properties of homogeneous-information equilibria*): *Under Assumption 1 and homogeneous information, there exist four symmetric equilibria in which changes in individual stock prices exhibit identical variance, Var , and identical correlation, $Corr$, between every pair of stocks with the following properties: As $\sigma_{\eta}^2 \rightarrow 0$,*

- (i) (*low volatility, low correlation*) $Var \searrow \sigma_{\delta}^2/r^2$, $Corr \rightarrow \rho_{\delta}$,
- (ii) (*high volatility, high correlation*) $Var \nearrow \infty$, $Corr \rightarrow 1$,
- (iii) (*high volatility, low correlation*) $Var \nearrow \infty$, $Corr \rightarrow \frac{-\rho_{\eta}}{1+(K-2)\rho_{\eta}}$,
- (iv) (*high volatility, negative correlation*) $Var \nearrow \infty$, $Corr \rightarrow -1/(K-1)$.

²⁰In a continuous-time model, Driessen, Maenhout, and Vilkov (2005) also assume a single instantaneous correlation between every pair of Wiener processes driving stock prices in their main analysis on correlation risk.

The first equilibrium is a low volatility, low correlation equilibrium. As the supply shocks become less volatile in this equilibrium, the common variance of changes in individual stock prices decreases. This occurs as the second moments of price changes become progressively dominated by dividend shocks; in the limit, the variance and correlation of price changes converge to σ_δ^2/r^2 and ρ_δ , respectively. These are the variance and correlation in a fixed-supply model: It is straightforward to show that, if the supply in our model were fixed at some constant N , with full information there would be a unique equilibrium with prices $\tilde{P}_t = r^{-1}\tilde{D}_{t+1} - r\bar{\theta}\Sigma_\delta^{-1}N$ and hence $Var(\Delta\tilde{P}_t) = \Sigma_\delta/r^2$. A similar result holds under no information with $\tilde{D}_{t+1}$ replaced by $\tilde{D}_t$.

In contrast, the common variance of individual stocks' price changes in the second equilibrium diverges to infinity as the supply shock variances fall. Moreover, the prices become perfectly correlated in the limit. We therefore call this equilibrium a high volatility, high correlation equilibrium. Strikingly, this occurs regardless of ρ_δ and ρ_η ; that is, high correlation obtains even though all the underlying shocks are uncorrelated or even negatively correlated. This is in sharp contrast to existing multisecurity rational expectations models that require *some* underlying correlation to produce equilibrium comovement (see, for example, Kodres and Pritsker (2002, Proposition 2) and Admati (1985, Section 5)). In the other two equilibria the common variance of individual stocks' price changes also diverges to infinity, while the correlation approaches some fixed number less than one. In one of these equilibria, the limiting correlation $\frac{-\rho_\eta}{1+(K-2)\rho_\eta}$ is close to zero if ρ_η is small or K is large (a high volatility, low correlation equilibrium). Its sign is the opposite of ρ_η .²¹ The other equilibrium has a limiting correlation $-1/(K-1)$ that is always negative and smaller than the limiting correlation of the third equilibrium (a high volatility, negative correlation equilibrium).²² Since we do not usually observe strongly negative return correlations in stock markets, the second and third equilibria seem empirically relevant given evidence on excessive volatility.²³

The varying equilibrium properties correspond to different beliefs that agents may have about the volatility of a set of mutual funds. Spiegel (1998) discusses how equilibrium prices can be "excessively" volatile in his no-information model even when stocks pay constant dividends. We extend his intuition and argue that strong *correlation* can occur as well if investors form beliefs about the price variability of portfolios rather than that of individual stocks. Assume that there are two symmetric securities with constant dividends and independent supply. Specifically, we set $K = 2$, $\sigma_\delta^2 = 0$, and $\rho_\delta = \rho_\eta = 0$ in Assumption 1. Since dividends are perfectly predictable, the full- and no-information models (as well as the partial-information model) coincide. This is also seen from the fact that setting $\Sigma_\delta = 0$ in equations

²¹This is so because $1 + (K-2)\rho_\eta > 0$ in equilibrium for all $K \geq 1$ and $-1 < \rho_\eta < 1$. The claim is immediate when $\rho_\eta \geq 0$. When $\rho_\eta < 0$, the positive definiteness of Σ_η guarantees that $[1 + (K-1)\rho_\eta]\sigma_\eta^2 > 0 > \rho_\eta\sigma_\eta^2$, where we note that the left-most term is an eigenvalue of Σ_η (see the proof of Proposition 2). Rearranging this inequality confirms the claim.

²²To see this, compute $\frac{-\rho_\eta}{1+(K-2)\rho_\eta} - \frac{-1}{K-1} = \frac{1-\rho_\eta}{[1+(K-2)\rho_\eta](K-1)} > 0$.

²³Note that Spiegel (1998) focuses on the first (low volatility, low correlation) and third (high volatility, low correlation) equilibria of his no-information model (see his Lemma 2 and calibration in Section 2).

(11) and (13) yields an identical quadratic matrix equation. Solving the equation, we obtain four (2^2) solutions for the A matrix of the form²⁴

$$A = \frac{\lambda_1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{\lambda_2}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (21)$$

where λ_1 and λ_2 each can take one of two values, 0 or $-r/\bar{\theta}\sigma_\eta^2$. The price function in (10) (or (12)) implies that the A matrix represents how supply shocks affect prices and hence investors' portfolio decisions. Alternatively, since the variance of price changes is linear in $-A$ (see equations (19) and (20)), it also represents agents' beliefs about the covariance structure of stock returns. Consider an economy in which investors believe that they can perfectly forecast future prices. Since they regard stocks as riskless assets, they will voluntarily provide perfectly elastic demand at prices $\frac{1}{r}D$, where D is the vector of sure dividends. This corresponds to an equilibrium in which $\lambda_1 = \lambda_2 = 0$, that is, $A = 0$. The zero loading on supply implies that investors do not price nonfundamental shocks (such as supply shocks) in this equilibrium.

However, if investors think that prices will be volatile, a different story emerges. Since they no longer regard stocks as risk-free assets, they require compensation for holding stocks. The larger the supply shocks, the higher the risk they must bear. This makes them require more compensation in the form of lower prices, which in turn implies that they will submit less elastic demand schedules, corresponding to the negative values of λ_1 and/or λ_2 . First, consider the case in which $\lambda_1 = -r/\bar{\theta}\sigma_\eta^2 < 0$ and $\lambda_2 = 0$. This produces a price function

$$\tilde{P}_t = -\frac{r}{2\bar{\theta}\sigma_\eta^2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \tilde{N}_{1t} \\ \tilde{N}_{2t} \end{pmatrix} + \frac{1}{r}D,$$

where $\tilde{N}_{1t}$ and $\tilde{N}_{2t}$ are the random supplies of the two stocks ($\tilde{N}_t = [\tilde{N}_{1t}, \tilde{N}_{2t}]^T$). Since the two stock prices are identical up to the constant dividend vector, they are perfectly correlated. Note that this occurs even though there is absolutely no correlation between the underlying shocks. This represents a highly volatile, strongly correlated equilibrium. If, instead, investors believe that $\lambda_1 = \lambda_2 = -r/\bar{\theta}\sigma_\eta^2 < 0$,

$$\tilde{P}_t = -\frac{r}{2\bar{\theta}\sigma_\eta^2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} \tilde{N}_{1t} \\ \tilde{N}_{2t} \end{pmatrix} + \frac{1}{r}D.$$

In this high volatility, low (zero) correlation equilibrium, the stock prices are uncorrelated and the two stock markets operate independently. Finally, the belief that $\lambda_1 = 0$ and $\lambda_2 = -r/\bar{\theta}\sigma_\eta^2 < 0$ produces a high volatility, negative correlation equilibrium.

²⁴Corollary 1 also holds when Σ_δ is zero, and hence positive semidefinite, as in this example. In this case A can be zero and thus negative semidefinite. See also footnote 7. We also note that since M is proportional to the identity matrix, any vector can serve as its eigenvector and therefore there are infinite equilibria. In this pathological case, we restrict the eigenvectors to those given in the proof of Proposition 2 in the Appendix.

What are λ_1 and λ_2 economically? They represent the variances of two uncorrelated mutual funds. These mutual funds are given by the eigenvectors of A , $x_1 \equiv \frac{1}{\sqrt{2}}[1 \ 1]^T$ and $x_2 \equiv \frac{1}{\sqrt{2}}[1 \ -1]^T$, corresponding to the two eigenvalues, λ_1 and λ_2 , respectively. The x_1 vector is an “equal-share” portfolio, which captures the movement in the aggregate stock market given the symmetry. The x_2 vector is a long-short portfolio in which the second stock is shorted to finance the purchase of the first.²⁵ The magnitude of λ_1 and λ_2 represents the variances of changes in the two portfolio values because the variance of price changes is linear in $-A$; for example, from equation (19) (or (20), with $\Sigma_\delta = 0$), the variance of the change in the value of portfolio x_1 is given by $x_1^T(-rA/\bar{\theta})x_1 = (r/\bar{\theta})(-\lambda_1)$.

D. Calibrating the Full-Information Model

This subsection examines whether our homogeneous-information models can fit stock return volatilities observed in the data. Since Spiegel (1998) calibrates his no-information model, we focus on the full-information model. We employ the simplest multisecurity economy with two symmetric securities, $K = 2$ in Assumption 1. The following example will be used throughout the rest of the paper.

EXAMPLE 1 (*Two symmetric securities*): *There are two securities with cross-sectionally identical dividend- and supply-shock variances*

$$\Sigma_\delta = \sigma_\delta^2 \begin{pmatrix} 1 & \rho_\delta \\ \rho_\delta & 1 \end{pmatrix}, \quad \Sigma_\eta = \sigma_\eta^2 \begin{pmatrix} 1 & \rho_\eta \\ \rho_\eta & 1 \end{pmatrix}. \quad (22)$$

The parameter values are taken from the empirical literature where possible. Given the overlapping generations structure, estimates with a relatively low frequency would be appropriate for the current model. Using ten-year time intervals, Shiller (1981b) finds the volatilities of the aggregate dividend shock and the aggregate price change at $\sigma_{\delta, Agg} = 16.5$ and $\sigma_{\Delta P, Agg} = 69.4$, respectively. Henceforth, we denote aggregate quantities with subscript *Agg* to distinguish them from individual ones. We begin by constructing a benchmark economy that fits Shiller’s estimates and evaluate how changes in parameter values alter equilibrium properties. Toward this end, we follow Spiegel (1998, Lemma 2) and assume that the aggregate supply $N = \frac{1}{2}[1 \ 1]^T$ and that the dividend shock correlation $\rho_\delta = 0$. Setting the aggregate dividend-shock volatility $N^T \Sigma_\delta N = \sigma_{\delta, Agg}^2$, we back out the individual dividend-shock volatility to be $\sigma_\delta = 23.3$. The interest rate is chosen somewhat arbitrarily at 5% per annum, or $R = 1.05$ ¹⁰. We set the individual supply-shock volatility $\sigma_\eta = 4.99 \times 10^{-3}$, which generates Shiller’s (1981b) aggregate volatility level in the analysis below on partial-information equilibria (see Section II.C). The average

²⁵Since prices are random, x_1 is neither equally nor value-weighted in “dollar” terms (in terms of the units of the consumption good). Similarly, x_2 is generally not a zero-investment portfolio. In the general case of asymmetric $K \geq 2$ securities without Assumption 1, we can still show that the uncorrelated mutual fund with the maximal variance involves no short selling as long as price changes between all stock pairs are positively correlated. This is an application of a mathematical result known as the Perron-Frobenius theorem.

risk-aversion parameter $\bar{\theta}$ is set at unity.²⁶ Since one can show that the volatility of the aggregate price change is a function of the product $\bar{\theta}\sigma_\eta$ (rather than $\bar{\theta}$ and σ_η separately) in a homogeneous-information equilibrium, this implies that the value of σ_η above can alternatively be interpreted as that of $\bar{\theta}\sigma_\eta$. The choice of Σ_ε and Σ_ζ is irrelevant in a homogeneous-information equilibrium and is deferred until the analysis of a partial-information equilibrium.

Figures 1 & 2
about here

Figures 1 and 2 plot the volatility and correlation, respectively, of changes in individual stock prices against ρ_δ and ρ_η .²⁷ In each figure Panel A represents a low volatility, low correlation equilibrium, in which the dividend shocks play a dominant role; note that the correlation in Panel A of Figure 2 is almost identical to the dividend shock correlation, ρ_δ (see the contour on the “ground”). This is consistent with Panel A of Figure 1, where the volatility of price changes at all points is only slightly higher than the fixed-supply limit, $\sigma_\delta/r = 37.1$ (see Proposition 2 (i)). Observe that volatility in the other three equilibria can be several times higher than in Panel A. As the middle expression in equation (19) implies, this disparity in the variance of price changes across different equilibria is due to varying contributions of the supply shock variance. Panel B represents a high volatility, high correlation equilibrium. The correlation in Panel B of Figure 2 is higher than 0.5 everywhere, including the origin. That is, strikingly, a strongly correlated equilibrium exists even when there is absolutely no underlying correlation. Panel C demonstrates the existence of a high volatility, low correlation equilibrium. As seen by the correlation in Panel C of Figure 2 being close to $-\rho_\eta$, supply shocks are almost the sole determinant of price characteristics in this equilibrium. Unless the two supply shocks are unrealistically extremely correlated, price correlation will be weak. The last panel depicts a high volatility, negative correlation equilibrium. The contours in Panels B through D of Figure 1 show that the high volatility in the last three equilibria is caused in large part by the supply shocks amplified by the A matrix.

Do these equilibria exist if prices are only partially revealing? What are the trading strategies of heterogeneously informed investors in such markets? These are the questions we now turn to.

II. Partial-Information Equilibria

A. Equilibrium Characterization

More than a quarter century ago, in his seminal work on rational expectations equilibrium, Grossman (1978, p.94) noted that “theorems of [perfect aggregation] are too strong to be true statements about the world.” Although his main point was on the stability of an equilibrium when information is costly, his remark also applies to our homogeneous-information equilibria. As seen in the previous section, while these equilibria may explain the excessive volatility and comovement observed in the data, they lead

²⁶Blume, Easley, and O’Hara (1994) and Wang (1994) also use a CARA parameter of one.

²⁷When $\rho_\delta = -\rho_\eta$, there are infinite equilibria since M is proportional to I , which admits any vector as its eigenvector. On such points, we restrict the eigenvectors to x_1 and x_2 in the previous subsection. Also see footnote 24.

to implications that the empirical literature has consistently rejected: (a version of) the CAPM and two-fund monetary separation. This is where we call for a partial-information equilibrium. Information asymmetry implies that investors hold diverse portfolios of risky assets. Since investors draw different mean-variance frontiers, although each of them holds a tangency portfolio that is efficient up to their individual information set, the market portfolio may not be efficient for any single investor.

The price function under partial information includes components similar to those under homogeneous information. The second term in equation (15) is a perpetuity paying $\tilde{D}_t$, and the first term is a discount from the fundamental value due to supply pressure (recall that A_1 is negative definite). An important difference from the case of homogeneous information is that the prices reveal noisy information, $\tilde{\xi}_t$, about future dividends. Here, the supply shocks serve as noises that prevent prices from fully revealing future dividends; knowing $\tilde{D}_t$ and $\tilde{N}_{t-1}$, agents can back out $\tilde{\xi}_t \equiv \tilde{\delta}_{t+1} + B_2^{-1} A_2 \tilde{\eta}_t$ from the prices, but not its two components $\tilde{\delta}_{t+1}$ and $\tilde{\eta}_t$ separately. In this way, market prices serve as noisy public signals. This is a standard feature of a noisy rational expectations equilibrium.

Given market prices, private signals, and endowments, each agent updates his posterior distribution of one-period-ahead dividends. Part (iii) of Theorem 1 states that agents' demands are a linear function of these conditioning variables. Unfortunately, the system of nonlinear matrix equations characterizing the equilibrium given in the Appendix does not admit an analytic solution. In the following subsections, we rely on numerical methods to analyze the properties of partial-information equilibria.²⁸

B. Stock Price Volatility in a Single-Security Model

We start the calibration with a single-security model ($K = 1$) with no information asymmetry, $\Sigma_{\varepsilon,i} \equiv \sigma_\varepsilon^2 \forall i$. We set the dividend shock variance equal to Shiller's (1981b) aggregate estimate, $\sigma_\delta^2 = \sigma_{\delta,Agg}^2$. To set the common private signal-error variance σ_ε^2 , we borrow from Cho and Krishnan (2000). Using S&P500 futures data, they estimate the average private signal-error volatility for Hellwig's (1980) single-security model at 20.705 over a 7-week horizon, with a dividend shock volatility of 5.495 (see their Table 2). Assuming serial independence of the private signal errors over time, we set the base value for σ_ε at $\sigma_{\varepsilon 0} = 20.705 \times 16.5 / 5.495 = 62.2$. Since no estimate is available for the variance of individual endowment noises, it is set somewhat arbitrarily at $\Sigma_\zeta^{1/2} \equiv 4\Sigma_\eta^{1/2}$ throughout the rest of the calibration.²⁹ The values

²⁸We only look for equilibria in which the coefficient matrices have spectral decompositions of the form (A27) in the Appendix and reduce the system of nonlinear matrix equations to a system of nonlinear scalar equations for eigenvalues similarly to (A28). Due to the lack of an analytic solution, it is not easy to derive conditions for the existence of a partial-information equilibrium. However, we see from Corollary 1 that when there exists a no-information equilibrium, there also exists a full-information equilibrium. Therefore, it appears reasonable to conjecture that a sufficient condition for the existence of a partial-information equilibrium is the existence of a no-information equilibrium with $\Sigma_{\varepsilon,i}^{-1} = 0 \forall i$ and otherwise identical parameter values.

²⁹Although the endowment can also be a variable of potential interest, we do not explore its informational role in this paper given our primary interest in the effect of diverse private information. Trial computations indicate that with the parameter values provided here, setting the endowment noise volatility to approximately four times the supply shock volatility

of r and $\bar{\theta}$ are retained from the previous section.

Figure 3 plots the volatility of the price change, $\sigma_{\Delta P}$, against the supply shock volatility, σ_{η} .³⁰ Five curves are shown in the figure. The left-most curve represents Spiegel's (1998) no-information equilibrium. In this equilibrium agents receive infinitely noisy (or simply no) private signals about future dividends. As we go from the left to the right, agents' private information becomes more accurate: The next three curves correspond to $\sigma_{\varepsilon} = \sigma_{\varepsilon 0} \times 1, 0.5$, and 0.25 , respectively. The right-most curve is the other extreme with perfect information, or the full-information equilibrium. The three partial-information equilibria reside between these two extreme cases. As we can see from the figure, for a given combination of supply shock volatility and private signal-error volatility, there are potentially two equilibria with differential levels of price variability (except for the knife-edge case at the right edge of a curve where these two equilibria coincide).³¹ In the low volatility equilibrium, a decrease in the supply shock volatility reduces the volatility of the price change, while the reverse is true in the high volatility equilibrium. Clearly in the latter, price variability can be excessive relative to dividend variability, since a very low supply-shock volatility can produce disproportionately high price variability.

The figure also depicts the effect of information. Holding the supply shock volatility constant, as the private information becomes more accurate the volatility of the price change falls in the low volatility equilibrium (as one goes down along the vertical axis in the lower limbs of the curves), while it rises in the high volatility equilibrium (as one goes up in the upper limbs). This effect is stronger at higher levels of supply shock volatility. In fact, the following corollary shows that a partial-information equilibrium (with multiple securities and asymmetric information) converges to a full- or no-information equilibrium as information becomes infinitely accurate ($\sigma_{\varepsilon} \rightarrow 0$) or noisy ($\sigma_{\varepsilon} \rightarrow \infty$).

COROLLARY 2 (*Convergence of a partial-information equilibrium*): *A partial-information equilibrium converges to a full-information equilibrium as $\Sigma_{\varepsilon,i} \rightarrow 0 \forall i$, or to a no-information equilibrium (when one exists) as $\Sigma_{\varepsilon,i}^{-1} \rightarrow 0 \forall i$.*

This formally confirms that a partial-information equilibrium resides between the full- and no-information equilibria. Therefore, as $\sigma_{\eta} \rightarrow 0$, the volatility of price changes in a partial-information equilibrium reaches the same limit as that in the corresponding homogeneous-information equilibrium stated in Proposition 2: The volatility diverges to ∞ in a partially revealing high volatility equilibrium, or converges to $\sigma_{\delta}/r = 16.5/0.63 = 26.2$ in a partially revealing low volatility equilibrium. We can see or higher produces virtually no difference in equilibrium quantities (except for increased trading volume), suggesting that its informational role is negligible at such values.

³⁰The moment expressions necessary for plotting this and subsequent figures are available in the technical appendix posted on the author's home page.

³¹Again, due to the lack of an analytic solution, it is difficult to pin down the number of partially revealing equilibria when they do exist. In the numerical methods, various starting values are examined. In the specific examples used in this paper, we numerically find two equilibria when $K = 1$ (single-security model), and four equilibria when $K = 2$ (two-security model).

Figure 3
about here

this property in Figure 3. The dashed line corresponds to Shiller's (1981b) aggregate volatility estimate, which is consistent with high rather than low volatility equilibria at all information levels. Point "A" represents the benchmark single-security economy that gives his volatility estimate with a private signal-error volatility implied by Cho and Krishnan (2000).

Figure 4
about here

Figure 4 shows how price properties vary with the common private signal-error volatility, σ_ε . Point "A" is again our benchmark economy. The circles and stars represent the high and low volatility equilibria, respectively. Panel A again confirms the opposing effects of information on the price variability in the two equilibria. Panel B plots the price sensitivity to the future dividend shock, B_2 . In both equilibria, the sensitivity increases with better information (moving us to the left) and converges to the full-information value, $1/r = 1.59$. This is the familiar multiple from the perpetuity formula for a sure payoff (which appears as the coefficient on $\tilde{D}_{t+1}$ in the full-information price formula (10)). The other limit is zero, because when agents receive infinitely noisy private information, there is no information for the price to aggregate in the first place; recall that in Spiegel's (1998) no-information model, the price function does not depend on the future dividend shock, $\tilde{\delta}_{t+1}$ (see equation (12)). Panel C shows that the absolute price sensitivity to the supply shock, $|A_2|$, behaves differently in the two equilibria. Note that the absolute value is plotted here as A_2 is a negative number. The shapes of the curves resemble those in Panel A especially for the upper one, confirming the role of the supply shock as the key determinant of price variability in the high volatility equilibrium.

Let us now examine whether existing stories (not necessarily mutually exclusive) can explain the opposing effects of information on volatility in the two equilibria (see, for example, Wang (1993, 1994) and West (1988)):

- (i) (Diminishing price discount: Volatility $\downarrow$) First, accurate information may reduce the price discount, as investors perceive less future uncertainty and thus require a lower premium to hold risky assets. Under this explanation, absolute price sensitivity to supply shocks will decline with information precision, and so will volatility.
- (ii) (Arbitrage trading: Volatility $\downarrow$) A second story suggests that trading of rational informed investors should always stabilize price variability, since such investors will take profitable positions whenever prices deviate from fundamental values. These trades will tend to pull prices back toward the "rational" values. Under this explanation, absolute loadings on supply shocks should again decrease with information accuracy.
- (iii) (Prices as aggregators of private information: Volatility $\downarrow$) In a noisy rational expectations framework, prices aggregate agents' noisy private signals. As private information becomes more accurate, prices will progressively reveal true dividends, and load more on dividend shocks than supply shocks. If the decrease in absolute supply-shock loadings outweighs the increase in dividend shock loadings, the volatility will decline.

- (iv) (Adverse selection: Volatility $\uparrow$) When information asymmetry is severe, informed trading might destabilize prices. In such a case, less informed traders will face an adverse selection problem and require a larger price discount to hold risky assets. This tends to make prices more sensitive to supply shocks.

Our low volatility equilibrium is perfectly consistent with the first three stories; the absolute supply-shock sensitivity decreases with better information (Panel C of Figure 4) while the dividend shock sensitivity rises (Panel B). Overall, the volatility of the price change falls (Panel A). However, the high volatility equilibrium is hard to explain because the absolute supply-shock sensitivity increases with information accuracy in Panel C.

The last story cannot explain the price behavior in our high volatility equilibrium either, since the result holds regardless of the degree of information asymmetry; recall that we have assumed that all investors are equally accurately informed in this calibration exercise ($\Sigma_{\varepsilon,i} \equiv \sigma_{\varepsilon}^2 \forall i$). The irrelevance of information asymmetry is another distinctive feature of our model.³² For example, price variability in Wang (1993, Figures 2 and 3), measured in terms of the price innovation or the price level, can increase with the fraction of informed investors. As he notes, however, the adverse selection problem plays an important role in his model, as there is almost always a strictly positive measure of uninformed investors.³³ Thus, the cause of the volatile price in the high volatility equilibrium must lie outside the traditional realm and is unique to our model. We argue that it is the self-fulfilling prophecies of overlapping generations supporting the amplified supply shock in the equilibrium price. In a high volatility equilibrium under partial revelation, investors hold the belief that a very small supply shock can produce a disproportionately large price variance. (If there are multiple stocks, these beliefs represent the variances of uncorrelated mutual funds in Section I.C.) By Corollary 2, an increase in information accuracy moves us toward the full-information equilibrium, which exhibits the most volatile price of all feasible equilibria for a given level of supply shock.

Finally, with noisy private signals, it is interesting to ask whether the multiplicity of equilibria is due to heterogeneous information or overlapping generations. The primary answer to this question is “the latter” (coupled with supply shocks), even though multiplicity is not uncommon in noisy rational expectations equilibrium models.³⁴ This is particularly evident from the fact that multiple equilibria

³²Lambert, Leuz, and Verrecchia (2006) make a similar point in connection to the cost of capital. They point out that it is the investors’ average information precision, not information asymmetry per se, that affects a firm’s cost of capital in a model with perfect competition. Their focus is on the first moment of returns, while we are primarily interested in the second moments.

³³This is also the approach taken in Biais, Bossaerts, and Spatt (2006). It would be suitable for their purpose to examine how uninformed investors should structure their portfolios. Because an econometrician’s information set is close to that of the uninformed investors, they are also able to test their model’s implication empirically. In contrast, the way we introduce heterogeneous information, based on Admati (1985), allows us to get rid of information asymmetry completely and analyze the pure effect of common information precision if desired.

³⁴Multiple partially revealing equilibria obtain, for example, in Grundy and McNichols (1989), Brown and Jennings

also exist under full or no information.

C. Price Volatility and Correlation with Multiple Risky Securities

In this subsection, we extend the analysis to the cross-sectional correlation between changes in stock prices under partial revelation. For this purpose, we examine a two-security benchmark economy with cross-sectionally independent private signal errors. Equating the aggregate private signal-error variance to Cho and Krishnan's (2000) estimate yields $\bar{\Sigma}_\varepsilon = 88.0^2 I$.

Panels A and B of Figure 5 plot the volatility and correlation, respectively, of changes in individual stock prices against σ_η . The results are identical for both securities because of symmetry. For a sufficiently low level of supply shock volatility, the panels show that, again, there exist four equilibria with the following properties:

- Equilibrium 1 (stars): low volatility, low (zero) correlation,
- Equilibrium 2 (squares): high volatility, high correlation,
- Equilibrium 3 (circles): high volatility, low (zero) correlation, and
- Equilibrium 4 (crosses): high volatility, negative correlation.

In all the figures to follow, the same marker is used to represent a particular equilibrium. Note that in Panels A and B, two equilibria, which differ across the two panels, are superimposed on the middle curve. Panel A shows that in the three high volatility equilibria the price variability increases with a decline in supply shock volatility, while it falls in the low volatility equilibrium. In Panel B, the two equilibria on the middle line (circles and stars) have zero correlation between changes in stock prices. In these equilibria, the two security markets operate independently. In contrast, in the two correlated equilibria (squares and crosses), the magnitude of correlation increases dramatically with a decline in supply shock volatility.

Since empirical studies may find positive dividend correlations, next we set $\rho_\delta = 0.3$, which implies that the individual dividend-shock volatility, σ_δ , equals 20.5 given Shiller's (1981b) aggregate figure. Other parameters are held unchanged. The results are shown in Panels C and D of Figure 5. In Panel C, we now see the four equilibria separately, three of which again exhibit higher price variability with less supply-shock volatility. Panel D shows that the correlation levels on the middle two curves are relatively weak. Throughout the four panels in the figure, Points A and B produce Shiller's (1981b) aggregate volatility estimate in the high volatility, high correlation equilibrium and the high volatility, low correlation equilibrium, respectively.

Several observations are worth noting. First, Points A and B achieve the same aggregate volatility level at a common individual supply-shock volatility despite the fact that individual stocks have lower

Figure 5
about here

price variability at Point A than at Point B (Panels A and C). This arises because the higher correlation at Point A (see Panels B and D) contributes to the aggregate volatility and makes up the deficiency in individual stocks' volatility. Second, comparing Panels A and B to Panels C and D reveals that the common supply-shock volatility at Points A and B is invariant to ρ_δ . This property obtains since we keep the aggregate dividend-shock variance constant. Finally, using a no-information model with cross-sectionally independent dividend and supply shocks, Spiegel (1998, Lemma 2) shows that the magnitude of supply shock variance necessary to reconcile a given level of aggregate price-change variance in a single-security model can be reduced by a factor of $1/K$ in a K -security model. The numerical result indicates that the common supply-shock volatility at Points A and B throughout the panels in Figure 5 is 4.99×10^{-3} . This equals $1/\sqrt{2}$ times 7.07×10^{-3} , the supply shock volatility in the single-security benchmark economy in Figure 3. Thus, we have numerically shown that Spiegel's claim also holds in our partial-information example. All the above three points can be proved analytically in a full- or no-information equilibrium.

We now analyze the effect of information on the properties of multisecurity partial-information equilibria. We keep the same parameter values at Points A and B in the independent dividend-shock example ($\rho_\delta = 0$) and change the average volatility of private signal errors. Figure 6 shows how the properties of individual stock prices vary with information precision. Panel A indicates that the common volatility of changes in individual stock prices ($\sigma_{\Delta P}$) increases with information quality in the three high volatility equilibria, while it decreases in the low volatility equilibrium. In Panel B, we see that the magnitude of correlation between the price changes ($\rho_{\Delta P}$) becomes even larger in the high and negative correlation equilibria (squares and crosses), while it is invariant at zero in the other two equilibria. Panel C shows that a security's price sensitivity to its own future dividend shock ($B_2(k, k)$, the k 'th diagonal element of B_2 , $k = 1$ or 2) increases with information quality in all four of the equilibria and converges to the full-information value, $1/r$. The other limit is zero, the no-information value. Again, this is because prices serve as aggregators of agents' private information. In Panel D, a stock's absolute price sensitivity to its own supply shock ($|A_2(k, k)|$, the absolute value of the k 'th diagonal element of A_2 , $k = 1$ or 2) rises with information accuracy in the high volatility, low correlation equilibrium (circles), while it falls in the low volatility equilibrium (stars). Interestingly, the relation is not monotone in the other two equilibria superimposed on the middle curve. Given the monotonicity of the price change volatility in Panel A, this suggests that at a relatively high level of information accuracy, the dividend shocks start contributing to the price variability in the high and negative correlation equilibria.³⁵

³⁵The fact that two equilibria are superimposed in each of these four panels crucially depends on the assumption of cross-sectionally independent dividend and supply shocks. In general cases including $\rho_\delta \neq 0$ and/or $\rho_\eta \neq 0$, the following limiting values are useful: As $\sigma_\varepsilon \rightarrow 0$ ($\sigma_\varepsilon \rightarrow \infty$), the volatility in Panel A converges to the square root of $Var_{FI,ind}$ ($Var_{NI,ind}$) and the correlation in Panel B to $Corr_{FI,ind}$ ($Corr_{NI,ind}$) in the proof of Proposition 2 in the Appendix. The Appendix shows how to choose the pair of eigenvalues λ_1 and λ_2 for these formulae in each of the four equilibria. The two limits of the price sensitivity to dividend shocks in Panel C do not depend on ρ_δ or ρ_η (see equations (10) and (12)).

Figure 6
about here

D. Trading Behavior of Asymmetrically Informed Agents

The rest of the paper analyzes the trade of asymmetrically informed agents under partial revelation. In the current model, there are three motives for agents to trade: Information asymmetry, random endowments, and taste (risk aversion). We devise volume measures that capture the first motive and examine the properties of these measures.

Since each agent is infinitesimally small, we consider the trading behavior of groups of agents. Divide the total mass of agents into J groups, each indexed by j with strictly positive measure, $m^j > 0$.³⁶ Groups here can be considered various investor classes. For example, they may represent individual and institutional investors, or domestic and foreign investors in an international context. We assume that the informational characteristic of each group is time invariant in the sense that the average variance, $\bar{\Sigma}_\varepsilon^j$, of private signal errors is the same over two successive generations of each group j .³⁷ Let $\Delta\tilde{\Pi}_{t,i} \equiv \Delta X_{t,i} - \tilde{\eta}_{t,i}$ be the net demand change, or the *net flow*, over two successive generations of agent i . Then, the net flows of group- j agents are

$$\Delta\tilde{\Pi}_t^j \equiv \int_{i \in j} \Delta\tilde{\Pi}_{t,i} di. \quad (23)$$

Following Brennan and Cao (1996, 1997), we define a measure of trading behavior as the covariance between group- j net flows and price changes, $Cov(\Delta\tilde{\Pi}_t^j, \Delta\tilde{P}_t^T)$, where $\Delta\tilde{P}_t \equiv \tilde{P}_t - \tilde{P}_{t-1}$. A positive diagonal element of this covariance matrix implies that group- j agents as a whole tend to purchase the corresponding security when its price has appreciated. That is, they behave like trend-followers. Conversely, if the covariance is negative, they follow a contrarian strategy, selling the security upon price appreciation. We now analyze this covariance matrix numerically.

For brevity, we present the results for a single-security economy. Results with two securities are available in the technical appendix posted on the author's home page; an advantage of such a multisecurity model is that it allows a richer cross-sectional information structure.³⁸ We use the same parameter values corresponding to the single-security benchmark economy in Figure 3 and vary the information precision. We assume that there are two groups of agents ($J = 2$) with equal measure and common average risk aversion $\bar{\theta}^1 = \bar{\theta}^2 = 1$. This ensures that there is no trade motive due to difference in risk aversion. To study the trading behavior of heterogeneously informed agents, we further assume that

From equation (A32), a stock's absolute price sensitivity to its own supply shock in Panel D converges to the same form $|\lambda_1 + (K - 1)\lambda_2|/K$ in the two limits, with $\lambda_1^\pm$ and $\lambda_2^\pm$ appropriately substituted.

³⁶Henceforth, a superscript denotes a group.

³⁷The formulae for $\bar{\Sigma}_\varepsilon^j$ and $\bar{\theta}^j$ below can be found in the proof of Theorem 1 in the Appendix.

³⁸For example, in an international context, domestic investors may be better informed about the domestic market portfolio, but less informed about a foreign one (see, for example, Brennan and Cao (1997)). A numerical analysis shows that in such a setting the domestic investors behave like contrarians on the domestic market portfolio, but like trend-followers on the foreign market portfolio. If one group is less informed about *all* securities than the other group, the analysis indicates that the former acts as trend-followers of the market portfolio, buying all securities upon price appreciation.

the two groups are asymmetrically informed, with the first group better informed on average about the stock than the second group. Specifically, we set $\bar{\Sigma}_\varepsilon^1 = 0.5\sigma_\varepsilon^2$. Setting the aggregate average variance of the private signal errors at $\bar{\Sigma}_\varepsilon = \sigma_\varepsilon^2$, this implies that the second group has $\bar{\Sigma}_\varepsilon^2 > \sigma_\varepsilon^2$.

Figure 7
about here

Figure 7 shows the correlation between the net flow of the first group and the price change, $\rho(\Delta\tilde{\Pi}_t^1, \Delta\tilde{P}_t)$. The correlation is negative in both equilibria. This implies that the better informed agents tend to sell the stock upon price appreciation, behaving like contrarians. The magnitude of the correlation is larger in the low volatility equilibrium (stars), because the price signal is relatively more informative than it is in the high volatility equilibrium.

To understand how agents in the second group trade, consider the three trade motives mentioned earlier. In group net flows, the effect of random endowments is void since the aggregation in (23) washes away the noise in individual endowments by the law of large numbers. Then, with common risk aversion, only information asymmetry is responsible for differential trading behavior. By market clearing, $\Delta\tilde{\Pi}_t^1 + \Delta\tilde{\Pi}_t^2 = 0$. It follows that $Cov(\Delta\tilde{\Pi}_t^1, \Delta\tilde{P}_t) = -Cov(\Delta\tilde{\Pi}_t^2, \Delta\tilde{P}_t)$. Thus, if one group acts as a contrarian, the other must behave like a trend-follower, which in our case is the second group. Intuitively, since the price is more informative to the less informed agents, they will rely on the public price signal more than the better informed investors and will trade in the same direction as price changes. Because $\rho(\Delta\tilde{\Pi}_t^2, \Delta\tilde{P}_t) = -\rho(\Delta\tilde{\Pi}_t^1, \Delta\tilde{P}_t)$, the graph for the second group is exactly the mirror image of the first group's and hence is omitted.

From the figure, as agents in the economy become more informed on average (moving us to the left), the contrarian behavior of the first group weakens and so does the trend-following behavior of the second group. This is because the partially revealing equilibria converge to the corresponding full-information equilibria with no information asymmetry (see Corollary 2).³⁹

The result on the trend-following behavior of less informed agents is similar to the findings in Brennan and Cao (1996, 1997) and Wang (1993). The current paper complements their work by demonstrating that such trading behavior can also occur in excessively volatile, possibly strongly correlated stock markets.

E. Trading Volume under Partial Revelation

Asymmetric information leads to strictly positive trading volume. In this subsection, we consider two measures of volume, one representing aggregate flows and the other reflecting individual trades. These

³⁹In the other limit when the average information level becomes very noisy, the correlations in Figure 7 seem to converge to certain nonzero values. This might appear counterintuitive since in a no-information equilibrium, agents should be effectively homogeneously uninformed and therefore should not trade. This is due to the normalized nature of the correlation measure. Intuition (correctly) suggests that expected absolute flows (to be introduced soon) and hence the standard deviation of flows will tend to zero (see Panel A of Figure 8). Since correlation is covariance divided by the relevant two standard deviations, both the numerator and the denominator of the correlation formula converge to zero. The particular information structure employed here keeps the ratio bounded away from zero.

two measures are motivated by possible empirical applications; the former would be more relevant when one works with aggregate trade data (e.g., Brennan and Cao (1997)), while the latter may be suitable for individual account data.

The first volume measure we analyze is based on the net flows introduced in the previous section. The per capita absolute net flow, $\tilde{U}_t$, is the absolute shares purchased (or, equivalently, sold) between groups,⁴⁰

$$\tilde{U}_t \equiv \frac{1}{2} \sum_{j=1}^J |\Delta \tilde{\Pi}_t^j|. \quad (24)$$

We call this measure the *absolute flow*. Since $\Delta \tilde{\Pi}_t^j$ has zero mean, the *expected absolute flow*, U , is given by

$$U \equiv \mathbf{E}[\tilde{U}_t] = \sum_{j=1}^J \sqrt{\frac{1}{2\pi} \text{diag}(\text{Var}(\Delta \tilde{\Pi}_t^j))}, \quad (25)$$

where $\text{diag}(\cdot)$ returns a vector carrying the diagonal elements of its argument matrix and $\sqrt{\cdot}$ is the elementwise square-root operator.⁴¹ As noted earlier, with common risk aversion this measure captures the trade motive due to asymmetric information only, since endowment noises cancel out in aggregation.

The second measure of trading volume is the standard one that aggregates individual absolute net flows, $\tilde{V}_t \equiv \frac{1}{2} \int_i |\Delta \tilde{\Pi}_{t,i}| di$. *Expected volume*, V , is given by

$$V \equiv \mathbf{E}[\tilde{V}_t] = \int_i \sqrt{\frac{1}{2\pi} \text{diag}(\text{Var}(\Delta \tilde{\Pi}_{t,i}))} di. \quad (26)$$

Because the volume is measured (the absolute value is taken) before aggregation in $\tilde{V}_t$, unlike the expected absolute flow, this measure will be nonzero due to heterogeneous endowments even if all agents are equally risk averse and have homogeneous information.

Figure 8 shows the two volume measures in the high (circles) and low (stars) volatility equilibria. Both of them have a hump-shaped relation with information accuracy.⁴² Intuitively, under full information, all agents are perfectly informed, and there is no information-based trade. In the other extreme case of no information, agents are homogeneously uninformed, and again there is no trade due to information asymmetry. At intermediate levels of information accuracy, strictly non-nil trade will arise. Again, Point A represents the benchmark economy that produces Shiller's (1981b) aggregate volatility level with Cho and Krishnan's (2000) estimate of aggregate private signal-error variance. Its location implies that improving information quality at Point A will raise trading volume.

⁴⁰The division by two corrects for the double counting of buys and sells.

⁴¹The expression follows from the well-known fact that, for a scalar normal random variable $\tilde{x} \sim N(0, \sigma^2)$, $\mathbf{E}|\tilde{x}| = \sqrt{2\sigma^2/\pi}$. This can easily be extended to a multivariate normal vector by straightforward computation.

⁴²Computing trading volume requires the specification of private signal-error variance for each individual agent. We have set it at the average level of the group that the agent belongs to. As $\sigma_\varepsilon \rightarrow 0$ or ∞ , expected trading volume approaches the limit, $\Sigma_\zeta^{1/2}/\sqrt{2\pi}$, where we recall that we have set $\Sigma_\zeta^{1/2} \equiv 4\Sigma_\eta^{1/2}$.

Figure 8
about here

F. Relation between Absolute Flows and Absolute Price Changes

Another measure of interest is the correlation between absolute flows and absolute price changes. As the next proposition states, with two groups we can explicitly sign the correlation regardless of the number of securities, K , and the distribution of the risk-aversion parameter across agents, θ_i . In particular, the correlation is always nonnegative. Define $\rho_{n,l} \equiv \text{Corr}(\Delta\tilde{\Pi}_t^1(n), \Delta\tilde{P}_t(l))$, the correlation between stock n 's net flow for the first group and stock l 's price change (which equals $-\text{Corr}(\Delta\tilde{\Pi}_t^2(n), \Delta\tilde{P}_t(l))$, the negative of the same correlation for the second group).

PROPOSITION 3 (*Correlation between absolute flows and absolute price changes*): *When there are two groups ($J = 2$), the absolute flow of stock n is nonnegatively correlated with the absolute price change of any stock l , $1 \leq n, l \leq K$. The correlation increases in $\rho_{n,l}$ and is given by*

$$\text{Corr}(\tilde{U}_t(n), |\Delta\tilde{P}_t(l)|) = \frac{2}{\pi - 2} \left[\sqrt{1 - \rho_{n,l}^2} + \rho_{n,l} \arcsin \rho_{n,l} - 1 \right] \geq 0. \quad (27)$$

The equality holds if and only if $\rho_{n,l} = 0$.

Karpoff (1987) and Gallant, Rossi, and Tauchen (1992) document that high trading volume tends to be associated with large absolute returns. The above proposition implies that such findings are the other side of investors' trend-following and contrarian behavior. To see this, note that $\rho_{n,n}$ (setting $l = n$) is proportional to the n 'th diagonal element of $\text{Cov}(\Delta\tilde{\Pi}_t^1, \Delta\tilde{P}_t) = -\text{Cov}(\Delta\tilde{\Pi}_t^2, \Delta\tilde{P}_t)$, our measure of trading behavior analyzed in Section II.D. Thus, if one of the two investor groups behaves like trend-followers, the other will act as contrarians on a security ($\rho_{n,n} \neq 0$) if and only if its absolute flow is strictly positively correlated with its absolute price change ($\text{Corr}(\tilde{U}_t(n), |\Delta\tilde{P}_t(n)|) > 0$).⁴³

Figure 9 shows the correlation between the absolute flow and the absolute price change in the single-security economy introduced earlier.⁴⁴ As Proposition 3 asserts, the correlation is positive in both equilibria, meaning that high volatility tends to be associated with large trades in either direction. The positive correlation diminishes with information accuracy as the economy approaches full revelation.

Figure 9
about here

III. Conclusion

Empirical studies document that various investor classes follow trend-chasing and contrarian strategies in both domestic and international markets. Many of these markets are found to exhibit excess volatility and, in some cases, strong comovements in asset returns. This paper is an attempt to explain these seemingly anomalous phenomena from a fully rational perspective. Using an overlapping generations model with information asymmetry and random asset supply, we first show that asset prices can be

⁴³This statement is general and we do not require information asymmetry here. In general, a group can act as trend-followers or contrarians ($\rho_{n,n} \neq 0$) for two reasons: Information asymmetry and risk aversion. These are two of the three trade motives remaining in group net flows $\Delta\tilde{\Pi}_t^j$ after aggregation.

⁴⁴Again, a two-security example is available in the technical appendix posted on the author's home page.

highly volatile relative to dividend variability. The model produces multiple equilibria that can exhibit strong or weak correlations between asset returns, even when asset supplies and future dividends are cross-sectionally uncorrelated. As is common in noisy rational expectations equilibrium models, prices serve as noisy public signals about future dividends because they aggregate agents' private signals. This leads to heterogeneous trading behavior across asymmetrically informed agents. Since less informed agents rely on price signals more than better informed agents, the former trade in the same direction as price changes and behave like trend-followers, while the latter act as contrarians. In addition, trading volume has a hump-shaped relation with the average level of information precision, because agents are effectively homogeneously informed or uninformed at extreme levels of information precision. Moreover, a security's absolute trade flow is positively correlated with its absolute price change in a market with trend-followers and contrarians. Accurate information increases the volatility and correlation of changes in stock prices in the highly volatile, strongly correlated equilibrium.

As this paper is a first investigation into the intersection of the overlapping generations literature and the noisy rational expectations equilibrium literature, there remain several interesting directions to explore. First, welfare issues are not addressed in the current paper. It can be shown that our partial-information equilibrium with asymmetric information precision is not Pareto efficient (see a relevant discussion in Brennan and Cao (1996)). In such a case, social welfare can be improved by introducing additional trading sessions or by introducing derivative securities (Brennan and Cao (1996, 1997) and Cao (1999)). Extending this work to look at these two mechanisms may be worth pursuing, as the real world has clearly implemented both of them.

Second, this study does not consider the dynamics of market prices and agents' trades. Information is necessarily short-lived in the current model. Endowing agents with longer lives would allow for analysis of long-lived information, which can have a richer impact on price and trade dynamics. It would also provide for a more natural interpretation of trading strategies and volume, which are currently defined over two successive generations of agents.

Third, the existence of multiple equilibria naturally raises the question of stability. Using a rational expectations model with random supply, Gennotte and Leland (1990) demonstrate that crashes can occur with relatively little selling. In their model, supply of a single risky asset rises as its price falls due to investors' hedging activity. Since the excess demand function can be backward bending, a small change in information signals can cause discontinuity in equilibrium. This is not the case in the full- and no-information equilibria of the current model. A stock's price sensitivity to its own supply shock will always be negative in these equilibria and hence the demand function will always be downward sloping. However, due to the lack of a closed-form solution, additional assumptions may be necessary to establish the stability of our partial-information equilibria.

Finally, related to the issue of stability and welfare are the following questions: *Why* do prices in the real world remain highly volatile and, in some cases, strongly correlated? Do people really not prefer

less volatile markets? Why do regulators' efforts to stabilize prices, such as the circuit breaker rule and market makers' smooth-quoting requirement, sometimes fail to work? It may be the case that once an equilibrium has been reached, it is hard to upset, even though it may not be Pareto optimal. Examples of such suboptimal but stable equilibria can be found in everyday life. A classic one is the prevalence of the QWERTY keyboard over the more efficient Dvorak keyboard (David (1985)).⁴⁵

⁴⁵Other examples include metric systems (the U.S. vs. the SI metric), personal computers (PC vs. Mac), operating systems (Windows, Mac OS, UNIX, Linux), currency systems (various currencies and the introduction of the euro), and various electronic-device formats (DVD-R vs. DVD+R, VHS vs. Beta, etc.). Some standards are more prevalent and stable than others, and some are almost extinct. See Besen and Farrell (1994) and Katz and Shapiro (1994) for more on this subject.

Appendix: Proofs

Proof of Theorem 1: Since showing sufficiency is straightforward, we derive only the necessary conditions.

(a) Full-information equilibrium. Guess that $A_1 = A_2 \equiv A$ and $B_1 = B_2 \equiv B$, and write the price function in (4) as $\tilde{P}_t = A\tilde{N}_t + B\tilde{D}_{t+1} + c$. Then the excess returns in (6) is $\tilde{Q}_{t+1} = A\tilde{\eta}_{t+1} + B\tilde{\delta}_{t+2} + \tilde{D}_{t+1} - r\tilde{P}_t$. With full information ($\tilde{\delta}_{t+1} \in \mathcal{F}_{t,i}$),

$$\begin{aligned} \text{Var}(\tilde{Q}_{t+1}|\mathcal{F}_{t,i}) &= A\Sigma_\eta A^\top + B\Sigma_\delta B^\top \equiv S, \\ \mathbf{E}[\tilde{Q}_{t+1}|\mathcal{F}_{t,i}] &= \tilde{D}_{t+1} - r\tilde{P}_t. \end{aligned} \quad (\text{A1})$$

Thus, the optimal demand function in (8) is given by $X_{t,i} = \frac{1}{\theta_i} S^{-1}(\tilde{D}_{t+1} - r\tilde{P}_t)$. The market-clearing condition in (9) can then be written as

$$\int_i \frac{1}{\theta_i} S^{-1}(\tilde{D}_{t+1} - r\tilde{P}_t) di = \tilde{N}_t.$$

Comparing the coefficients on both sides of the equation gives $\frac{1}{\theta} S^{-1}(-rA) = I$, $I - rB = 0$, and $c = 0$. Rearranging the first condition and substituting equation (A1) for S with $B = \frac{1}{r}I$, we obtain $-\frac{r}{\theta}A = A\Sigma_\eta A^\top + \Sigma_\delta/r^2$. Since the right-hand side of this last equation is symmetric and positive definite, A is symmetric and negative definite. Dropping the transposition superscript gives the quadratic matrix equation for A in the theorem. Then using these conditions for the price coefficients, the demand function above reduces to

$$X_{t,i} = \frac{1}{\theta_i} S^{-1}(-rA\tilde{N}_t) = \frac{\bar{\theta}}{\theta_i} \tilde{N}_t.$$

(b) No-information equilibrium. This case is similar to the full-information equilibrium above and hence is omitted. Also see Spiegel (1998).

(c) Partial-information equilibrium. Using the assumed price function (4), we can write the excess return function (6) in terms of independent variables in the information set $\mathcal{F}_{t,i}$:

$$\tilde{Q}_{t+1} = (A_1 A_2^{-1} - I)B_2 \tilde{\xi}_t + A_2 \tilde{\eta}_{t+1} + G \tilde{\delta}_{t+1} + B_2 \tilde{\delta}_{t+2} + \tilde{D}_t - r\tilde{P}_t, \quad (\text{A2})$$

where

$$\begin{aligned} \tilde{\xi}_t &\equiv \tilde{\delta}_{t+1} + F\tilde{\eta}_t, \\ F &\equiv B_2^{-1}A_2, \end{aligned} \quad (\text{A3})$$

$$G \equiv B_1 + I - A_1 F^{-1}, \quad (\text{A4})$$

and we have assumed the nonsingularity of A_2 and B_2 . From normal updating theory, the conditional variance and mean of future dividends given $\mathcal{F}_{t,i}$ are

$$\text{Var}^{-1}(\tilde{\delta}_{t+1}|\mathcal{F}_{t,i}) = \Sigma_\delta^{-1} + (F^\top)^{-1}(\Sigma_\eta^{-1} + \Sigma_\zeta^{-1})F^{-1} + \Sigma_{\varepsilon,i}^{-1} \equiv \Sigma_i^{-1}, \quad (\text{A5})$$

$$\mathbf{E}[\tilde{\delta}_{t+1}|\mathcal{F}_{t,i}] = \Sigma_i[(F^\top)^{-1}\Sigma_\eta^{-1}F^{-1}\tilde{\xi}_t + \Sigma_{\varepsilon,i}^{-1}\tilde{z}_{t,i} + (F^\top)^{-1}\Sigma_\zeta^{-1}F^{-1}\tilde{\phi}_{t,i}] \equiv \tilde{\mu}_{t,i}, \quad (\text{A6})$$

where

$$\tilde{\phi}_{t,i} \equiv \tilde{\xi}_t - F\tilde{\eta}_{t,i} = \tilde{\delta}_{t+1} - F\tilde{\zeta}_{t,i}$$

represents the signals about future dividends inferred from the individual endowment and the price signals. From (A2), and noting that $\tilde{\delta}_{t+2}, \tilde{\eta}_{t+1} \notin \mathcal{F}_{t,i}$,

$$\text{Var}(\tilde{Q}_{t+1}|\mathcal{F}_{t,i}) = A_2\Sigma_\eta A_2^\text{T} + G\Sigma_i G^\text{T} + B_2\Sigma_\delta B_2^\text{T} \equiv S_i, \quad (\text{A7})$$

$$\mathbf{E}[\tilde{Q}_{t+1}|\mathcal{F}_{t,i}] = (A_1 A_2^{-1} - I)B_2\tilde{\xi}_t + G\tilde{\mu}_{t,i} + \tilde{D}_t - r\tilde{P}_t \equiv m_{t,i}. \quad (\text{A8})$$

The demand function in (8) is $X_{t,i} = \frac{1}{\theta_i} S_i^{-1} m_{t,i}$. Then the market-clearing condition (9) is

$$\int_i \frac{1}{\theta_i} S_i^{-1} m_{t,i} di = \tilde{N}_t. \quad (\text{A9})$$

Define average measures $\overline{S}$, $\overline{\Sigma}$, and $\overline{\Sigma}_\varepsilon$ by

$$(\overline{\theta S})^{-1} \equiv \int_i (\theta_i S_i)^{-1} di, \quad (\text{A10})$$

$$(\overline{\theta S})^{-1} G \overline{\Sigma} \equiv \int_i (\theta_i S_i)^{-1} G \Sigma_i di, \quad (\text{A11})$$

$$\overline{\Sigma}_\varepsilon^{-1} \equiv \overline{\Sigma}^{-1} - \Sigma_\delta^{-1} - (F^\text{T})^{-1} (\Sigma_\eta^{-1} + \Sigma_\zeta^{-1}) F^{-1}. \quad (\text{A12})$$

Comparing the coefficients in both sides of (A9) yields the following nonlinear system of matrix equations:

- (i) Coefficients on $\tilde{D}_t$: $\int_i (\theta_i S_i)^{-1} (I - rB_1) di = 0$, or

$$B_1 = \frac{1}{r} I. \quad (\text{A13})$$

- (ii) Coefficients on $\tilde{N}_{t-1}$:

$$\int_i (\theta_i S_i)^{-1} (-rA_1) di = I, \quad (\text{A14})$$

or

$$A_1 = -\frac{1}{r} \overline{\theta S}. \quad (\text{A15})$$

From this equation, A_1 is symmetric negative definite since S_i , and hence $\overline{S}$, are symmetric positive definite.

- (iii) Coefficients on $\tilde{\delta}_{t+1}$: Canceling the noise terms in $\tilde{z}_{t,i}$ and $\tilde{\phi}_{t,i}$, and using (A5),

$$\int_i (\theta_i S_i)^{-1} [(A_1 A_2^{-1} - I)B_2 + G\Sigma_i (\Sigma_i^{-1} - \Sigma_\delta^{-1}) - rB_2] di = 0. \quad (\text{A16})$$

Using the definitions in (A3) and (A11), we obtain

$$(\overline{\theta S})^{-1} [A_1 F^{-1} - B_2 + G - G\overline{\Sigma} \Sigma_\delta^{-1} - rB_2] = 0. \quad (\text{A17})$$

Here, by equations (A4) and (A13), we have

$$A_1 F^{-1} + G = \frac{R}{r} I. \quad (\text{A18})$$

Substituting equation (A18) into (A17) and solving for B_2 , it follows that

$$B_2 = \frac{1}{r}I - \frac{1}{R}G\bar{\Sigma}\Sigma_\delta^{-1}. \quad (\text{A19})$$

Note that there are other equivalent expressions.

(iv) Coefficients on $\tilde{\eta}_t$: Similarly,

$$\int_i (\theta_i S_i)^{-1} [(A_1 A_2^{-1} - I) B_2 F + G \Sigma_i (F^T)^{-1} \Sigma_\eta^{-1} F^{-1} F - r A_2] di = I, \quad (\text{A20})$$

or

$$A_1 - A_2 + G\bar{\Sigma}(F^T)^{-1}\Sigma_\eta^{-1} - rA_2 = \bar{\theta}\bar{S} = -rA_1,$$

where we use equations (A3) and (A15). Solving for A_2 , we obtain

$$A_2 = A_1 + \frac{1}{R}G\bar{\Sigma}(F^T)^{-1}\Sigma_\eta^{-1}. \quad (\text{A21})$$

(v) The constant terms: It is easy to see that $c = 0$.

The coefficient matrices A_1 , A_2 , and B_2 are a solution to the system of nonlinear matrix equations (A15), (A19), and (A21), with F , G , $\bar{S}$, and $\bar{\Sigma}$ defined in (A3), (A4), (A10), and (A11). Finally, using equations (A6) and (15), we may rewrite equation (A8) as $m_{t,i} = c_{0,i} + C_{1,i}\tilde{\xi}_t + C_{2,i}\tilde{z}_{t,i} + C_{3,i}\tilde{\phi}_{t,i} + C_{4,i}\tilde{D}_t + C_{5,i}\tilde{N}_{t-1}$ for some constant matrices $C_{1,i}, \dots, C_{5,i}$ and vector $c_{0,i}$. That is, given the normality assumption, the demand function is linear in the conditioning variables. In comparing the coefficients on $\tilde{\eta}_t$ in step (iv) above, we have started with $\int_i \frac{1}{\theta_i} S_i^{-1} C_{1,i} di \cdot F = I$. This implies that F , and hence A_2 and B_2 , must be nonsingular in equilibrium as assumed.

The group average measures, $\bar{\theta}^j$, $\bar{S}^j$, $\bar{\Sigma}^j$, and $\bar{\Sigma}_\varepsilon^j$, in Section II.D are defined analogously as the aggregate average measures:

$$(\bar{\theta}^j)^{-1} \equiv \frac{1}{m^j} \int_{i \in j} \theta_i^{-1} di, \quad (\text{A22})$$

$$(\bar{\theta}^j \bar{S}^j)^{-1} \equiv \frac{1}{m^j} \int_{i \in j} (\theta_i S_i)^{-1} di, \quad (\text{A23})$$

$$(\bar{\theta}^j \bar{S}^j)^{-1} G \bar{\Sigma}^j \equiv \frac{1}{m^j} \int_{i \in j} (\theta_i S_i)^{-1} G \Sigma_i di, \quad (\text{A24})$$

$$(\bar{\Sigma}_\varepsilon^j)^{-1} \equiv (\bar{\Sigma}^j)^{-1} - \Sigma_\delta^{-1} - (F^T)^{-1}(\Sigma_\eta^{-1} + \Sigma_\zeta^{-1})F^{-1}. \quad \text{Q.E.D.} \quad (\text{A25})$$

Proof of Corollary 1: Denote a symmetric positive-definite square root of Σ_η by $\Sigma_\eta^{\frac{1}{2}}$ such that $(\Sigma_\eta^{\frac{1}{2}})^2 = \Sigma_\eta$. Start with the quadratic matrix equation (11) under full information. Pre- and post-multiply $\Sigma_\eta^{\frac{1}{2}}$ to obtain

$$Y^2 + \frac{r}{\bar{\theta}}Y + \frac{1}{r^2}\Sigma_\eta^{\frac{1}{2}}\Sigma_\delta\Sigma_\eta^{\frac{1}{2}} = 0,$$

where $Y \equiv \Sigma_\eta^{\frac{1}{2}} A \Sigma_\eta^{\frac{1}{2}}$. Completing the square, we obtain $\left(Y + \frac{r}{2\theta} I\right)^2 = M_{FI}$, where M_{FI} is given by equation (18). Solving for Y and then for A yields

$$A = -\frac{r}{2\theta} \Sigma_\eta^{-1} + \Sigma_\eta^{-\frac{1}{2}} M_{FI\pm}^{\frac{1}{2}} \Sigma_\eta^{-\frac{1}{2}}, \quad (\text{A26})$$

where $M_{FI\pm}^{\frac{1}{2}}$ is a square root, not necessary positive definite, of M_{FI} . The whole set of $M_{FI\pm}^{\frac{1}{2}}$ is given by $M_{FI\pm}^{\frac{1}{2}} = C \Lambda_\pm^{\frac{1}{2}} C^T$, where C is the matrix of orthonormal eigenvectors of M_{FI} and $\Lambda_\pm^{\frac{1}{2}}$ is a diagonal matrix containing signed square roots of the corresponding eigenvalues $\lambda_1, \dots, \lambda_K$, that is, a matrix with elements $\pm\sqrt{\lambda_1}, \pm\sqrt{\lambda_2}, \dots, \pm\sqrt{\lambda_K}$ on the main diagonal with their signs freely chosen. It can be shown that the set of solutions in (A26) is unchanged if we take a square root of Σ_η that is not positive definite in the very first step. It is straightforward to show that the solution under no information is given by replacing M_{FI} with M_{NI} . This completes the proof. Q.E.D.

Proof of Proposition 1: Denote a portfolio by x , whose elements represent the number of shares held. From equation (19), the variance of changes in portfolio value under homogeneous information is given by

$$x^T \text{Var}(\Delta \tilde{P}_t) x = -\frac{r}{\theta} x^T A x = \frac{r}{\theta} \left(\frac{r}{2\theta} x^T \Sigma_\eta^{-1} x - x^T \Sigma_\eta^{-\frac{1}{2}} C \Lambda_\pm^{\frac{1}{2}} C^T \Sigma_\eta^{-\frac{1}{2}} x \right),$$

where we have substituted the solution for the A matrix in (17). The first term in the parentheses does not depend on the choice of equilibrium. Write $C^T \Sigma_\eta^{-\frac{1}{2}} x \equiv y$. Then the second term is $y^T \Lambda_\pm^{\frac{1}{2}} y = \sum_{l=1}^K (\pm\sqrt{\lambda_l} y_l^2)$, where y_l is the l 'th element of y and λ_l is the l 'th diagonal element of Λ . As one switches the sign on any $\sqrt{\lambda_l}$ from positive to negative, this quantity decreases and therefore the portfolio variance increases unless $y_l = 0$. Q.E.D.

Proof of Proposition 2:

(a) Full-information equilibrium. Under Assumption 1, it is straightforward to confirm that the K eigenvectors of Σ_δ and Σ_η are given by $x_1 = [1, \dots, 1]^T / \sqrt{K}$ and

$$x_m = \frac{1}{\sqrt{(m-1)m}} \left[\underbrace{1, \dots, 1}_{m-1 \text{ ones}}, \underbrace{-(m-1)}_{m\text{'th position}}, \underbrace{0, \dots, 0}_{K-m \text{ zeros}} \right]^T, \quad 2 \leq m \leq K.$$

The corresponding eigenvalues for Σ_δ are given by $\lambda_{\delta 1} = \sigma_\delta^2 [1 + (K-1)\rho_\delta]$ for x_1 and $\lambda_{\delta m} = \sigma_\delta^2 (1 - \rho_\delta)$ for x_m , $2 \leq m \leq K$, and similarly for Σ_η . Collect the eigenvectors in $C = [x_1, \dots, x_K]$ and the corresponding eigenvalues in diagonal matrices Λ_δ and Λ_η . Then the spectral decomposition of Σ_δ and Σ_η can be written as $\Sigma_\delta = C \Lambda_\delta C^T$, $\Sigma_\eta = C \Lambda_\eta C^T$. While it is possible to proceed with the general solution for the A matrix in Corollary 1, we exploit the symmetry assumption here. Guess that A also has the spectral decomposition

$$A = C \Lambda_A C^T = \sum_{m=1}^K \lambda_m x_m x_m^T, \quad (\text{A27})$$

where Λ_A is a diagonal matrix with eigenvalue λ_m on its m 'th main diagonal. Then the quadratic matrix equation (11) under full information is given as

$$C[\Lambda_\eta \Lambda_A^2 + \frac{r}{\bar{\theta}} \Lambda_A + \frac{1}{r^2} \Lambda_\delta] C^\top = 0. \quad (\text{A28})$$

Since C is nonsingular, this is equivalent to equating the terms inside the square bracket to zero. Because all the matrices involved are diagonal matrices of eigenvalues, this amounts to solving the following quadratic scalar problem for each eigenvalue:

$$\lambda_{\eta m} \lambda_m^2 + \frac{r}{\bar{\theta}} \lambda_m + \frac{1}{r^2} \lambda_{\delta m} = 0, \quad m = 1, \dots, K. \quad (\text{A29})$$

For $m = 1$, the two solutions to equation (A29) are

$$\lambda_1^\pm = \frac{-r\bar{\theta}^{-1} \pm \sqrt{r^2\bar{\theta}^{-2} - 4r^{-2}\sigma_\delta^2\sigma_\eta^2[1 + (K-1)\rho_\eta][1 + (K-1)\rho_\delta]}}{2\sigma_\eta^2[1 + (K-1)\rho_\eta]} < 0 \quad (\text{A30})$$

as long as Σ_δ and Σ_η are both positive definite. For $m = 2, \dots, K$,

$$\lambda_2^\pm = \lambda_m^\pm = \frac{-r\bar{\theta}^{-1} \pm \sqrt{r^2\bar{\theta}^{-2} - 4r^{-2}\sigma_\delta^2\sigma_\eta^2(1 - \rho_\eta)(1 - \rho_\delta)}}{2\sigma_\eta^2(1 - \rho_\eta)} < 0. \quad (\text{A31})$$

We focus on the case in which $\lambda_2 = \lambda_3 = \dots = \lambda_K$. As we will see below, this corresponds to symmetric equilibria. Decompose the spectral decomposition of A in (A27) into two parts, one representing the equal-share portfolio ($m = 1$) and another representing the long-short portfolios ($m \geq 2$). It can be verified that

$$A = \frac{\lambda_1}{K} \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \dots & \dots & 1 \end{pmatrix} + \frac{\lambda_2}{K} \begin{pmatrix} K-1 & -1 & \dots & -1 \\ -1 & K-1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & K-1 \end{pmatrix}. \quad (\text{A32})$$

From equation (A32) it is clear that changing the sign of one or more λ_m , $m \geq 2$, will produce an asymmetric equilibrium with cross-sectional variation in variance and correlation (unless all the signs are changed). Recall from (19) that $Var_{FI}(\Delta \tilde{P}_t) = -\frac{r}{\bar{\theta}} A$. Thus, the variance of any individual stock's price change is

$$Var_{FI,ind} = -\frac{r}{\bar{\theta}} \cdot \frac{\lambda_1 + (K-1)\lambda_2}{K} > 0.$$

The cross-sectional covariance between changes in any two stocks' prices is

$$Cov_{FI,ind} = -\frac{r}{\bar{\theta}} \cdot \frac{\lambda_1 - \lambda_2}{K}$$

and therefore the correlation is

$$Corr_{FI,ind} = \frac{\lambda_1 - \lambda_2}{\lambda_1 + (K-1)\lambda_2}.$$

Note that as $\sigma_\eta^2 \rightarrow 0$: $\lambda_1^- \rightarrow -\infty$, $\lambda_1^+ \rightarrow -r^{-3}\bar{\theta}\sigma_\delta^2[1 + (K-1)\rho_\delta]$, $\lambda_2^- \rightarrow -\infty$, and $\lambda_2^+ \rightarrow -r^{-3}\bar{\theta}\sigma_\delta^2(1 - \rho_\delta)$.

The four equilibria are characterized by the following sets of eigenvalues:

- (i) (low volatility, low correlation) $(\lambda_1, \lambda_2, \dots, \lambda_K) = (\lambda_1^+, \lambda_2^+, \dots, \lambda_2^+)$,
- (ii) (high volatility, high correlation) $(\lambda_1, \lambda_2, \dots, \lambda_K) = (\lambda_1^-, \lambda_2^+, \dots, \lambda_2^+)$,
- (iii) (high volatility, low correlation) $(\lambda_1, \lambda_2, \dots, \lambda_K) = (\lambda_1^-, \lambda_2^-, \dots, \lambda_2^-)$, and
- (iv) (high volatility, negative correlation) $(\lambda_1, \lambda_2, \dots, \lambda_K) = (\lambda_1^+, \lambda_2^-, \dots, \lambda_2^-)$.

Considering the limit of $Var_{FI,ind}$ and $Corr_{FI,ind}$ in each equilibrium gives the result in the proposition.

(b) No-information equilibrium. The eigenvalue problem in (A29) is replaced with

$$\lambda_{\eta m} \lambda_m^2 + \frac{r}{\theta} \lambda_m + \frac{R^2}{r^2} \lambda_{\delta m} = 0, \quad m = 1, \dots, K. \quad (A33)$$

Note that the only difference is the constant term. For $m = 1$, the solutions to equation (A33) are

$$\lambda_1^\pm = \frac{-r\bar{\theta}^{-1} \pm \sqrt{r^2\bar{\theta}^{-2} - 4R^2r^{-2}\sigma_\delta^2\sigma_\eta^2[1 + (K-1)\rho_\eta][1 + (K-1)\rho_\delta]}}{2\sigma_\eta^2[1 + (K-1)\rho_\eta]} < 0, \quad (A34)$$

and for $m = 2, \dots, K$, the solutions are

$$\lambda_2^\pm = \lambda_m^\pm = \frac{-r\bar{\theta}^{-1} \pm \sqrt{r^2\bar{\theta}^{-2} - 4R^2r^{-2}\sigma_\delta^2\sigma_\eta^2(1 - \rho_\eta)(1 - \rho_\delta)}}{2\sigma_\eta^2(1 - \rho_\eta)} < 0. \quad (A35)$$

By the formula (20) for the variance of price changes in a no-information equilibrium, we have the following expressions for the moments of individual stocks' price changes:

$$\begin{aligned} Var_{NI,ind} &= -\frac{r}{\theta} \cdot \frac{\lambda_1 + (K-1)\lambda_2}{K} - \frac{R^2 - 1}{r^2} \sigma_\delta^2 > 0, \\ Cov_{NI,ind} &= -\frac{r}{\theta} \cdot \frac{\lambda_1 - \lambda_2}{K} - \frac{R^2 - 1}{r^2} \sigma_\delta^2 \rho_\delta, \\ Corr_{NI,ind} &= \frac{Cov_{NI,ind}}{Var_{NI,ind}}. \end{aligned}$$

It is straightforward to verify that the limits of $Var_{NI,ind}$ and $Corr_{NI,ind}$ as $\sigma_\eta^2 \rightarrow 0$ are identical to those under full information in all four equilibria. Q.E.D.

Proof of Corollary 2:

(a) Convergence to the full-information equilibrium. When $\Sigma_{\varepsilon,i} \rightarrow 0 \forall i$, the conditional dividend-shock variance $Var(\tilde{\delta}_{t+1}|\mathcal{F}_{t,i}) \equiv \Sigma_i \rightarrow 0$ because private signals perfectly reveal future dividends. Thus, from equation (A7), the conditional variance of excess returns $Var(\tilde{Q}_{t+1}|\mathcal{F}_{t,i}) \equiv S_i \rightarrow A_2 \Sigma_\eta A_2^\top + B_2 \Sigma_\delta B_2^\top$. By the definitions in (A10) and (A11), the average measures $\bar{S}$ and $\bar{\Sigma}$ converge to the same limit as S_i and Σ_i , respectively, due to information homogeneity. Assuming G and F are finite, equations (A19) and (A21) then imply that $B_2 \rightarrow \frac{1}{r}I$ and $A_2 \rightarrow A_1$. Using these limits, equation (A15) converges to $A_1 \rightarrow -\frac{1}{r}\bar{\theta}[A_1 \Sigma_\eta A_1^\top + \Sigma_\delta/r^2]$. This is the full-information quadratic matrix equation in (11). Finally,

by definition, $F \rightarrow rA_1$ and $G \rightarrow I$ with $B_1 = \frac{1}{r}I$. Since A_1 is finite when the quadratic matrix equation has a real solution, so are F and G , as assumed.

(b) Convergence to the no-information equilibrium. Similar to the full-information case. When $\Sigma_{\varepsilon,i}^{-1} \rightarrow 0 \forall i$, private signals reveal no information about future dividends. While investors do have information about the aggregate supply shocks inferred from their own endowments, this information has no value because the investors can infer the realized aggregate supply shocks anyway from their own demands in equilibrium; recall that even in Spiegel's (1998) no-information model, the equilibrium demand (14) is proportional to $\tilde{N}_t$. Therefore, $Var(\tilde{\delta}_{t+1}|\mathcal{F}_{t,i}) \equiv \Sigma_i \rightarrow \Sigma_\delta = \bar{\Sigma}$. Then equation (A5) implies that $F^{-1} \rightarrow 0$. Assuming that A_2 is finite, the definition of F then requires that $B_2 \rightarrow 0$. So, from equation (A7), $Var(\tilde{Q}_{t+1}|\mathcal{F}_{t,i}) \equiv S_i \rightarrow A_2 \Sigma_\eta A_2^\top + G \Sigma_\delta G^\top = \bar{S}$. Further assuming that A_1 is finite, by definition $G \rightarrow (R/r)I$ with $B_1 = \frac{1}{r}I$. Then, by equation (A21), $A_2 \rightarrow A_1$. Applying these limits to equation (A15), we have $A_1 \rightarrow -\frac{1}{r}\bar{\theta}[A_1 \Sigma_\eta A_1^\top + (R^2/r^2)\Sigma_\delta]$. This is the no-information quadratic matrix equation in (13). Finally, both A_1 and A_2 are indeed finite in the limit when the quadratic matrix equation has a real solution. Q.E.D.

Proof of Proposition 3: When there are two groups of agents ($J = 2$), market-clearing implies that net trades occur strictly between them, that is, $|\Delta \tilde{\Pi}_t^1| = |\Delta \tilde{\Pi}_t^2|$. Thus, $\tilde{U}_t = (|\Delta \tilde{\Pi}_t^1| + |\Delta \tilde{\Pi}_t^2|)/2 = |\Delta \tilde{\Pi}_t^1| = |\Delta \tilde{\Pi}_t^2|$. The quantity of interest is $Corr(|\Delta \tilde{\Pi}_t^1(n)|, |\Delta \tilde{P}_t(l)|)$, where we have used $\tilde{U}_t = |\Delta \tilde{\Pi}_t^1|$. Observe that the two variables inside the absolute value operators are normally distributed, and the correlation can be calculated from the relevant noncentral moment; it is known that when $\tilde{x}$ and $\tilde{y}$ are bivariate standard normal variables with correlation ρ , $\mathbf{E}[\tilde{x}\tilde{y}] = 2(\sqrt{1-\rho^2} + \rho \arcsin \rho)/\pi$ (see, for example, Johnson and Kotz (1972)). Substituting this into the relation $Cov(|\tilde{x}|, |\tilde{y}|) = \mathbf{E}[\tilde{x}\tilde{y}] - \mathbf{E}[\tilde{x}]\mathbf{E}[\tilde{y}]$ with $\mathbf{E}[\tilde{x}]\mathbf{E}[\tilde{y}] = 2/\pi$ and dividing both sides by $\sqrt{Var(|\tilde{x}|)Var(|\tilde{y}|)} = (\pi - 2)/\pi$ gives equation (27) in the main text. Next, using the fact that

$$\frac{\partial \arcsin \rho}{\partial \rho} = \frac{1}{\sqrt{1-\rho^2}},$$

one can rewrite equation (27) as

$$Corr(|\Delta \tilde{\Pi}_t^1(n)|, |\Delta \tilde{P}_t(l)|) = \frac{2}{\pi - 2} \int_0^{\rho_{n,l}} \arcsin \rho d\rho \geq 0$$

for $-1 \leq \rho_{n,l} \leq 1$. The equality holds if and only if $\rho_{n,l} = 0$. Q.E.D.

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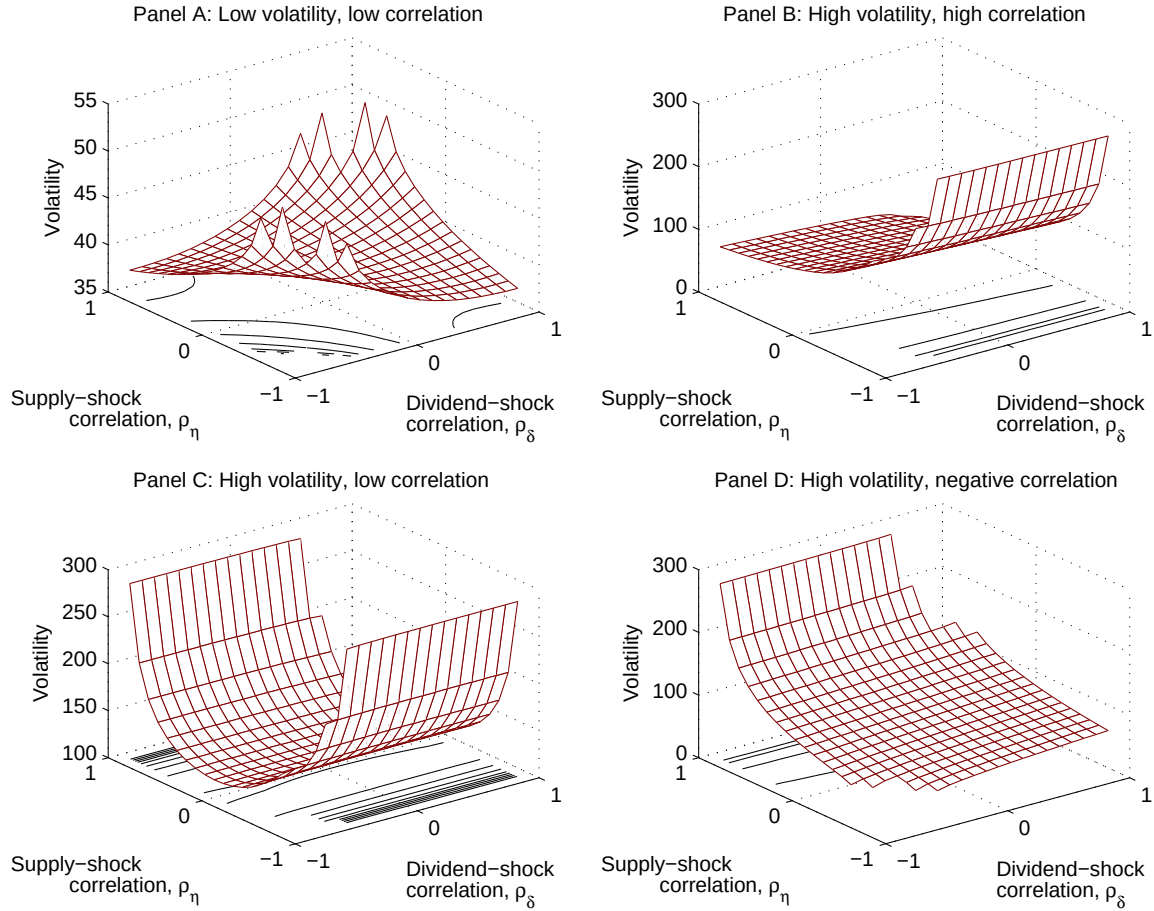


Figure 1: Volatility of individual stocks' price changes in a full information model with two symmetric securities. The panels represent the following four equilibria: Panel A—low volatility, low correlation; Panel B—high volatility, high correlation; Panel C—high volatility, low correlation; and Panel D—high volatility, negative correlation. Parameter values: $\Sigma_\delta = 23.3^2 \begin{pmatrix} 1 & \rho_\delta \\ \rho_\delta & 1 \end{pmatrix}$, $\Sigma_\eta = 0.00499^2 \begin{pmatrix} 1 & \rho_\eta \\ \rho_\eta & 1 \end{pmatrix}$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

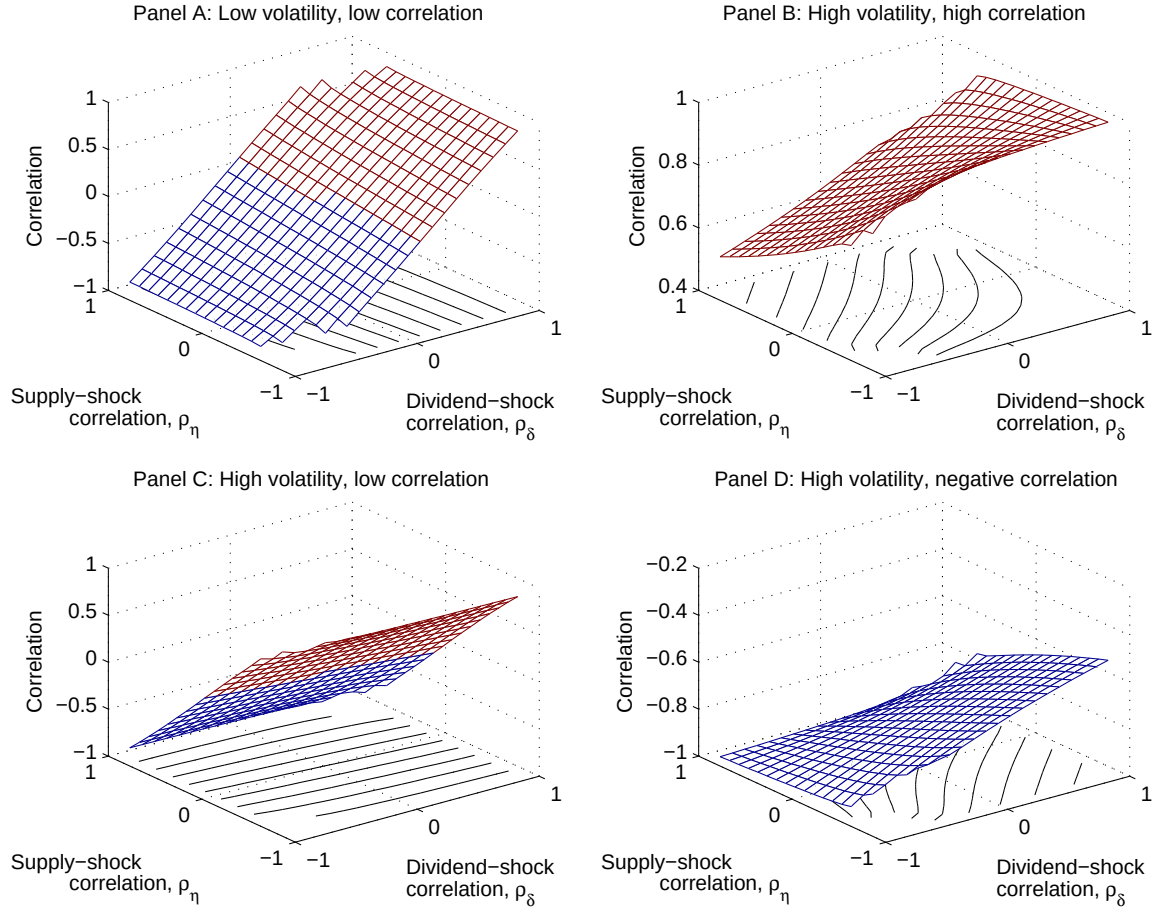


Figure 2: Cross-sectional correlation between individual stocks' price changes in a full information model with two symmetric securities. The panels represent the following four equilibria: Panel A–low volatility, low correlation; Panel B–high volatility, high correlation; Panel C–high volatility, low correlation; and Panel D–high volatility, negative correlation. Parameter values: $\Sigma_\delta = 23.3^2 \begin{pmatrix} 1 & \rho_\delta \\ \rho_\delta & 1 \end{pmatrix}$,

$$\Sigma_\eta = 0.004992 \begin{pmatrix} 1 & \rho_\eta \\ \rho_\eta & 1 \end{pmatrix}, \text{ and } r = 5\% \text{ per annum or } 1.05^{10} - 1.$$

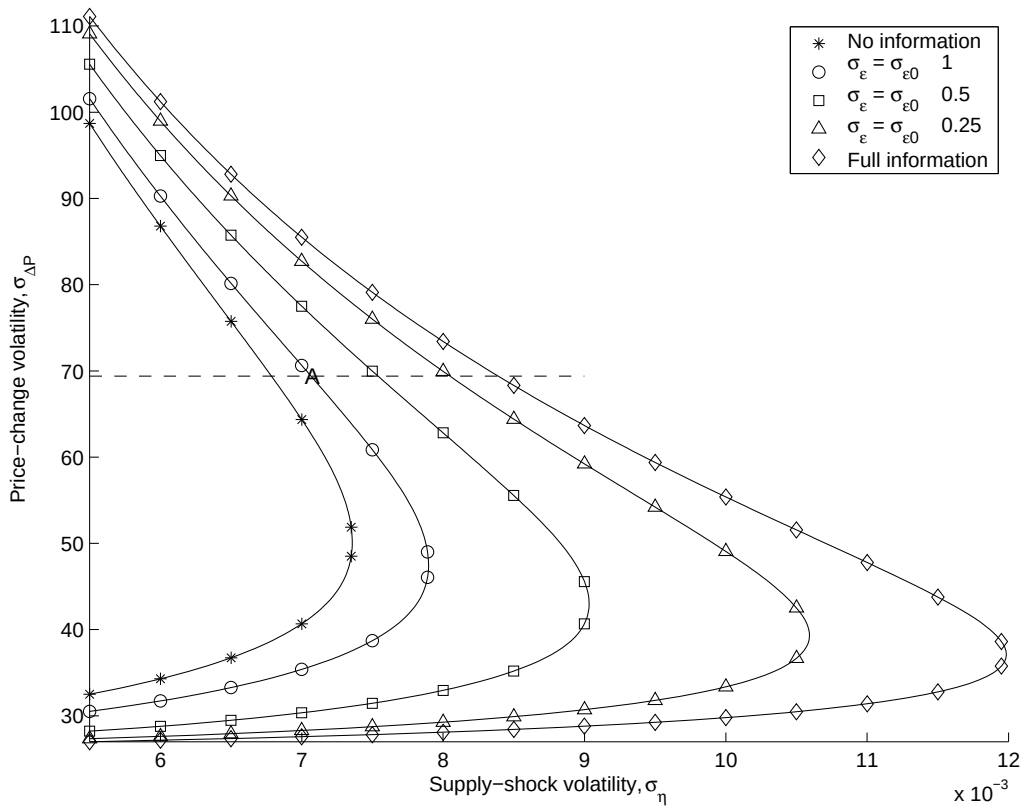


Figure 3: Volatility of the price change in a single-security model. The dashed line corresponds to Shiller's (1981b) aggregate volatility estimate, 69.4. Point A gives this volatility level under $\sigma_\varepsilon = \sigma_{\varepsilon 0} \equiv 62.2$. Parameter values: $\sigma_\delta = 16.5$, $\Sigma_\zeta^{1/2} = 4\sigma_\eta$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

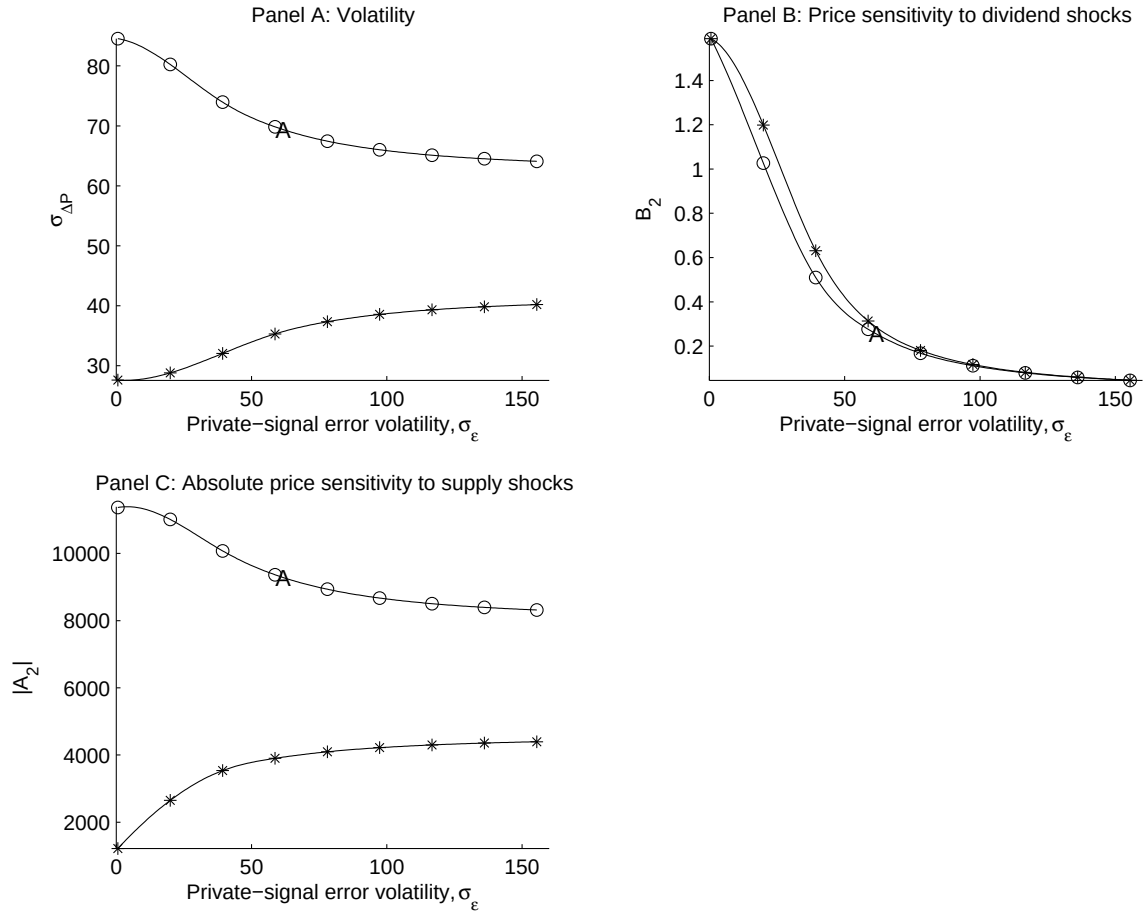


Figure 4: Price properties in a partial-information model with a single security. Panel A: Volatility of the price change, $\sigma_{\Delta P}$. Panel B: Price sensitivity to the future dividend shock, B_2 . Panel C: Absolute price sensitivity to the supply shock, $|A_2|$. The circles and stars represent high and low volatility equilibria, respectively. Point A gives Shiller's (1981b) aggregate volatility estimate, 69.4, at $\sigma_\epsilon = \sigma_{\epsilon 0} \equiv 62.2$. Parameter values: $\sigma_\delta = 16.5$, $\sigma_\eta = .00707$, $\Sigma_\zeta^{1/2} \equiv 4\sigma_\eta$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

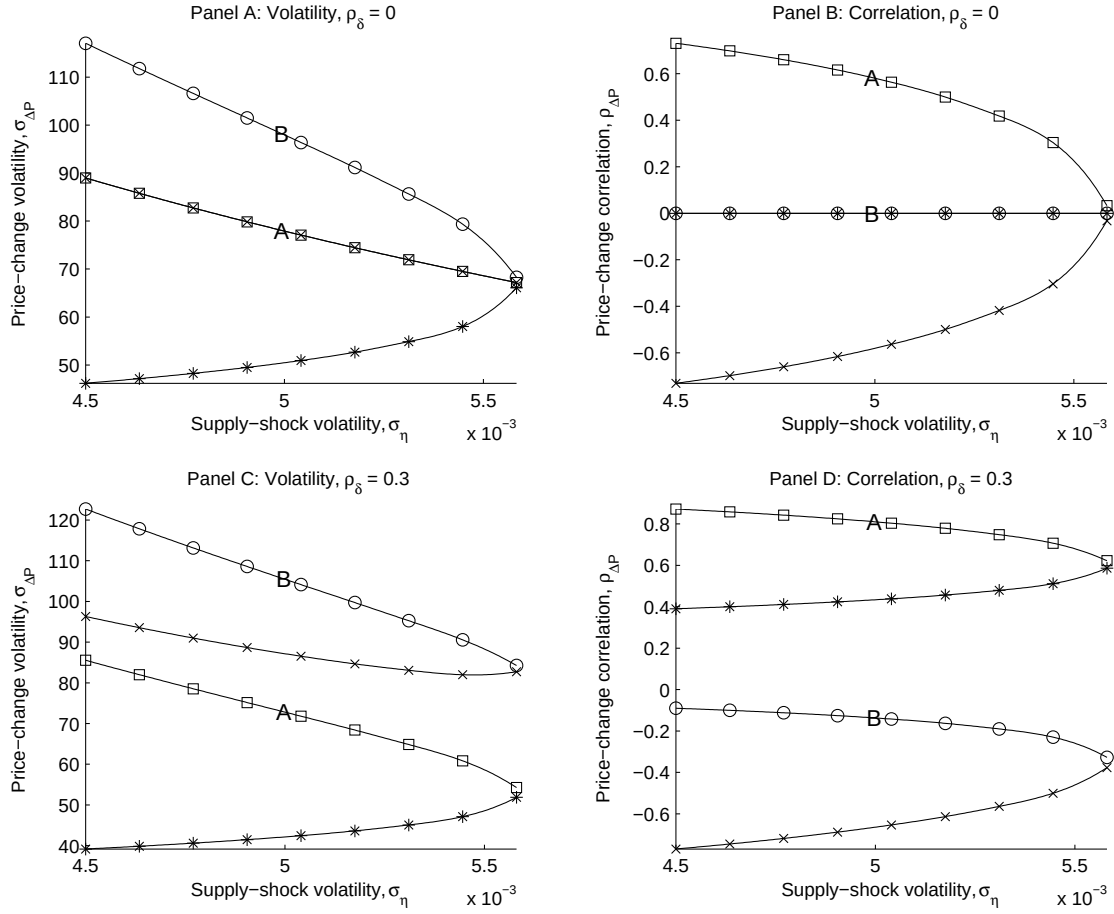


Figure 5: Volatility and correlation in a partial-information model with two symmetric securities. Panels A and C show the common volatility of individual stocks' price changes. Panels B and D plot the cross-sectional correlation between the two stocks' price changes. Panels A and B set the dividend shock correlation at $\rho_\delta = 0$, while Panels C and D set it at $\rho_\delta = 0.3$. The markers represent the following equilibria: stars—low volatility, low correlation; squares—high volatility, high correlation; circles—high volatility, low correlation; and crosses—high volatility, negative correlation. Points A and B give Shiller's (1981b) aggregate volatility estimate, 69.4. Parameter values: $\Sigma_\delta = \sigma_\delta^2 \begin{pmatrix} 1 & \rho_\delta \\ \rho_\delta & 1 \end{pmatrix}$ with $\sigma_\delta = 23.3$ in Panels A and B and $\sigma_\delta = 20.5$ in Panels C and D, $\Sigma_\eta = \sigma_\eta^2 I$, $\bar{\Sigma}_\varepsilon = 88.0^2 I$, $\Sigma_\zeta^{1/2} = 4\Sigma_\eta^{1/2}$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

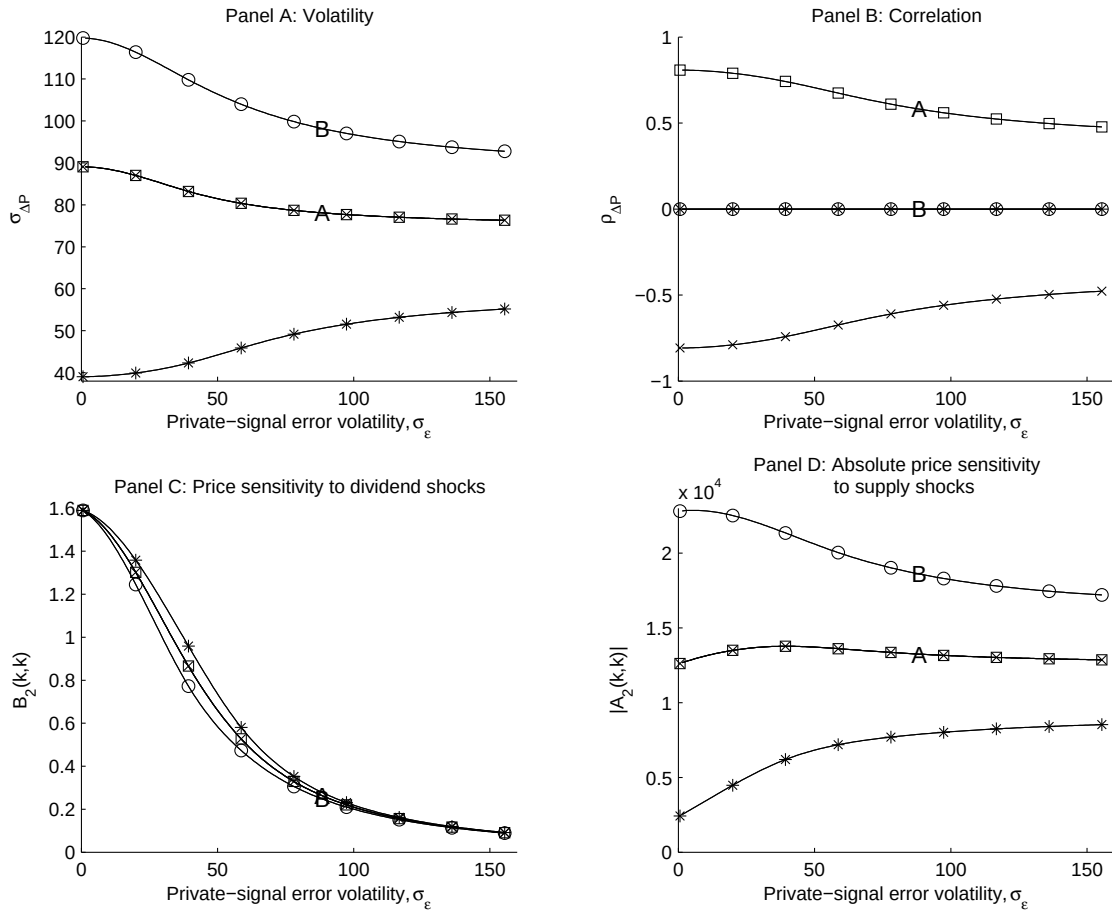


Figure 6: Price properties in a partial-information model with two symmetric securities. Panel A: Common volatility of individual stocks' price changes, $\sigma_{\Delta P}$. Panel B: Cross-sectional correlation between the two stocks' price changes, $\rho_{\Delta P}$. Panel C: A stock's price sensitivity to its own future dividend shock, $B_2(k, k)$. Panel D: A stock's absolute price sensitivity to its own supply shock, $|A_2(k, k)|$. The markers represent the following equilibria: stars—low volatility, low correlation; squares—high volatility, high correlation; circles—high volatility, low correlation; and crosses—high volatility, negative correlation. Points A and B give Shiller's (1981b) aggregate volatility estimate, 69.4. Parameter values: $\Sigma_\delta = 23.3^2 I$, $\Sigma_\eta = 0.00499^2 I$, $\bar{\Sigma}_\epsilon = \sigma_\epsilon^2 I$, $\Sigma_\zeta^{1/2} = 4\Sigma_\eta^{1/2}$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

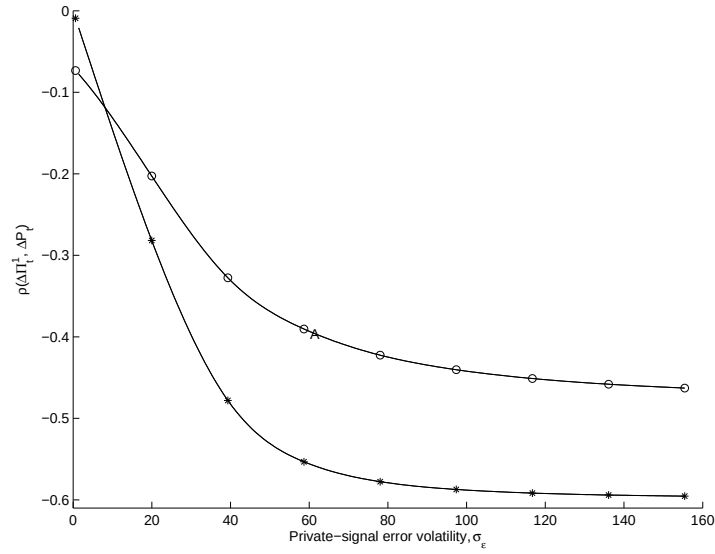


Figure 7: Trading behavior of asymmetrically informed agents in a partial-information model with a single security. There are two groups of agents. Group-1 agents are on average better informed about the stock than group-2 agents in that $\bar{\Sigma}_\varepsilon^1 = 0.5\sigma_\varepsilon^2$. The figure shows the correlation between the net flow of group-1 agents, $\Delta\tilde{\Pi}_t^1$, and the price change, $\Delta\tilde{P}_t$. The circles and stars represent the high and low volatility equilibria, respectively. Point A gives Shiller's (1981b) aggregate volatility estimate, 69.4, at $\sigma_\varepsilon = \sigma_{\varepsilon 0} \equiv 62.2$. Parameter values: $\sigma_\delta = 16.5$, $\sigma_\eta = 0.00707$, $\Sigma_\zeta^{1/2} = 4\sigma_\eta$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

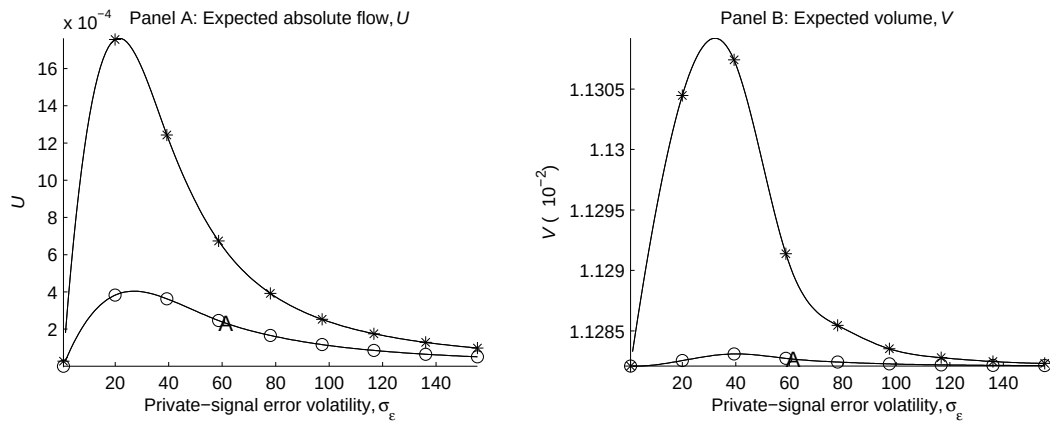


Figure 8: Trading volume between asymmetrically informed agents in a partial-information model with a single security. There are two groups of agents. Group-1 agents are on average better informed about the stock than group-2 agents in that $\bar{\Sigma}_\epsilon^1 = 0.5\sigma_\epsilon^2$. Panel A: The expected absolute flow, U . Panel B: Expected volume, V . The circles and stars represent the high and low volatility equilibria, respectively. Point A gives Shiller's (1981b) aggregate volatility estimate, 69.4, at $\sigma_\epsilon = \sigma_{\epsilon 0} \equiv 62.2$. Parameter values: $\sigma_\delta = 16.5$, $\sigma_\eta = 0.00707$, $\Sigma_\zeta^{1/2} = 4\sigma_\eta$, and $r = 5\%$ per annum or $1.05^{10} - 1$.

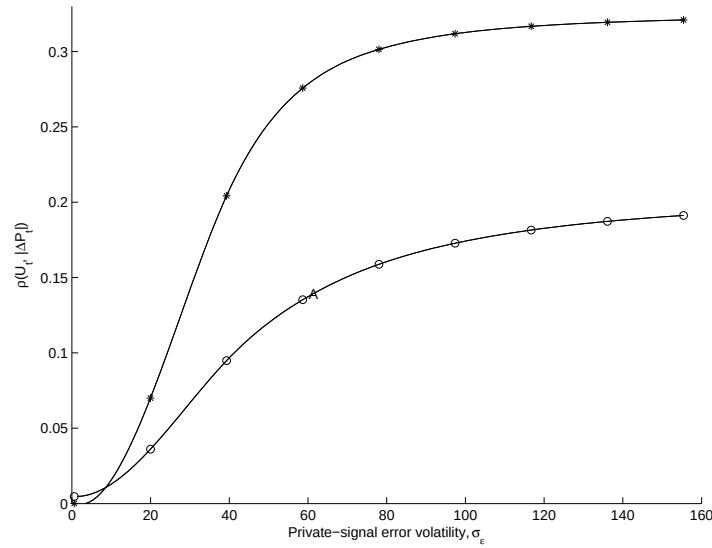


Figure 9: Correlation between the absolute flow, $\tilde{U}_t$, and the absolute price change, $|\Delta \tilde{P}_t|$, in a partial-information model with a single security. There are two groups of agents. Group-1 agents are on average better informed about the stock than group-2 agents in that $\bar{\Sigma}_\varepsilon^1 = 0.5\sigma_\varepsilon^2$. The circles and stars represent the high and low volatility equilibria, respectively. Point A gives Shiller's (1981b) aggregate volatility estimate, 69.4, at $\sigma_\varepsilon = \sigma_{\varepsilon 0} \equiv 62.2$. Parameter values: $\sigma_\delta = 16.5$, $\sigma_\eta = 0.00707$, $\Sigma_\zeta^{1/2} = 4\sigma_\eta$, and $r = 5\%$ per annum or $1.05^{10} - 1$.